

Language Models and RNNs

Lecture Plan

Today, we're diving in several **foundational concepts and tasks**.

1. Introduction to language modeling
2. Recurrent neural networks (RNNs)
3. Machine translation (a very language-y task w/ RNNs)

Intro to language modeling

Are these sentences okay?

Jane went to the store.

store to Jane went the.

Jane went store.

Consider the grammar
of the language

Jane goed to the store.

Consider morphology and exceptions

The store went to Jane.

Consider semantic categories

The food truck went to Jane.

and their exceptions

Language modeling

Language modeling is the task of predicting what word comes next

The students opened their ----

books

laptops

exams

minds

Given a sequence of words $x_1, x_2, \dots, x_t$, compute the probability distribution of the next word x_{t+1} :

$$P(x_{t+1} \mid x_t, \dots, x_1)$$

where x_{t+1} can be any word in the vocabulary.

Language modeling

For example, if we have some text

The students opened their laptops

$$P(\text{The, students, opened, their, laptops})$$

Remember **the chain rule!**

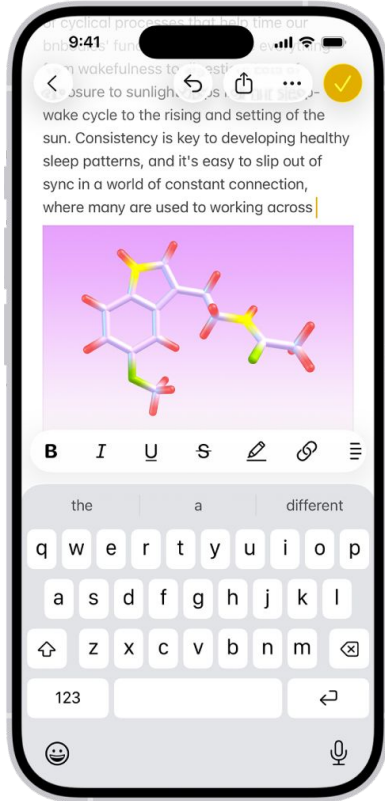
$$P(A, B) = P(B | A)P(A)$$

$$= P(\text{The}) \times P(\text{students} | \text{The}) \times \cdots \times P(\text{laptops} | \text{their, opened, students, The})$$

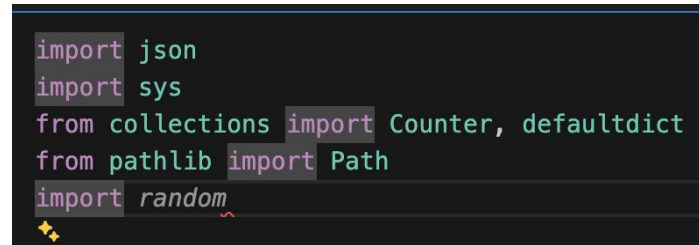
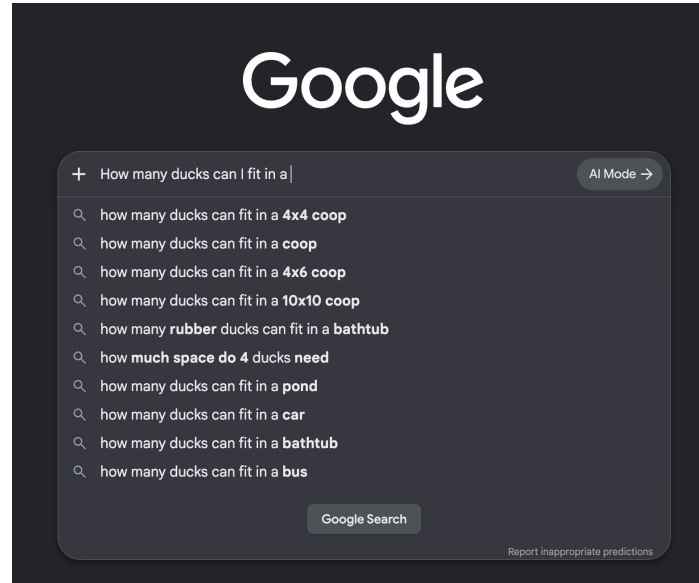
$$= \prod_{t=1}^T P(x_t | x_{t-1}, \dots, x_1)$$

Our LM provides us
these probabilities

Everyday examples



Predictive text; tap a suggestion to apply.



What can we do with LMs?

Score the probability of sentences

Jane went to the store. → high

store to Jane went the. → low

(same as calculating loss for training)

Generate sentences

while didn't choose end-of-sentence token (e.g. [EOS]):

calculate probability $P(\text{next word} \mid \text{context})$

sample a new word from the probability distribution

Language modeling

Language modeling is a **benchmark task** that helps us measure our progress on predicting language use

It's a **subcomponent** of many NLP tasks, especially those involving generation or estimating text probabilities:

predictive typing, speech recognition, handwriting recognition, spelling/grammar correction, authorship identification, machine translation, summarization, dialogue, etc ...

Everything in NLP is now built as a **language modeling** task

Everyday examples

 Happy Wednesday, Lucy

|How can I help you today?



Chat

Cowork

Fable 5 High



Why?

A strong language model is strong at lots of intelligent tasks.

With next word prediction, we can do:

Madison has a population of _____ people. [knowledge]

I put ___ fork down on the table. [syntax]

The woman walked across the street, checking for traffic over ___ shoulder. [coreference]

I went to the ocean to see the fish, turtles, seals, and ____. [lexical semantics/topic]

Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was ____. [sentiment]

The chef dropped the shrimp into the fryer, and watched it ____. [commonsense reasoning]

I was thinking about the sequence that goes 1, 1, 2, 3, 5, 8, 13, 21, ____. [math]

How did we do this before deep learning?

The students opened their ____

Learn an n -gram language model!

Unigrams : “the”, “students”, “opened”, “their”

Bigrams: “the students”, “students opened”, “opened their”

Trigrams: “the students opened”, “the students opened their”

4-grams: “the students opened their”

Main idea: collect statistics about how frequent different n -grams are and use these to predict the next word.

Unigram LM

Independence assumption: $P(x_i | x_1, \dots, x_{i-1}) \approx P(x_i)$

MLE of the dataset, where $c(x)$ is the count of word x

$$P(x_i) = \frac{c(x_i)}{\sum_{\tilde{x}} c(\tilde{x})}$$

Unigram LM

A unigram LM trained on Shakespeare:

To him swallowed confess hear both . Which . Of save on
trail for are ay device and rote life have

Hill he late speaks ; or ! a more to leg less first you
enter

Higher-order n-gram LMs

Markov assumption:

Predicting the next word only depends on the previous $n-1$ words.

$$P(x_i | x_{i-1}, \dots, x_1) \approx P(x_i | x_{i-1}, \dots, x_{i-n+1})$$

For example, for a **bigram LM**:

$$P(x_i | x_{i-1}, \dots, x_1) \approx P(x_i | x_{i-1})$$

To calculate these probabilities, count them in some large dataset:

$$P(x_i | x_{i-1}, \dots, x_{i-n+1}) \approx \frac{\text{count}(x_{i-n+1} \dots x_{i-2} x_{i-1} x_i)}{\text{count}(x_{i-n+1} \dots x_{i-2} x_{i-1})}$$

Bigram LM example

Given a training corpus of three sentences:

<s> I am Sam </s>

<s> Sam I am </s>

<s> I do not like green eggs and ham </s>

Compute bigram probabilities by counting:

$$P(I | <s>) = \frac{2}{3} = 0.67$$

$$P(\text{Sam} | <s>) = \frac{1}{3} = 0.33$$

$$P(\text{am} | I) = \frac{2}{3} = 0.67$$

$$P(</s> | \text{Sam}) = \frac{1}{2} = 0.50$$

$$P(\text{Sam} | \text{am}) = \frac{1}{2} = 0.50$$

$$P(\text{do} | I) = \frac{1}{3} = 0.33$$

Probability of a sentence = product of all bigram probabilities in the sentence.

$$P(<s> I \text{ am Sam } </s>) = \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{9} = 0.\bar{1}$$

A comparison of n -gram LMs

A **unigram** LM trained on Shakespeare:

To him swallowed confess hear both . Which . Of save on
trail for are ay device and rote life have

Hill he late speaks ; or ! a more to leg less first you
enter

A **4-gram** LM trained on Shakespeare:

King Henry. What! I will go seek the traitor Gloucester.
Exeunt some of the watch. A great banquet serv'd in;

It cannot be but so.

So who even uses n -gram LMs today?

Not very many people, but ...

Infini-gram ([Liu et al. 2024](#)) is a **modernized** n -gram LM, or a ∞ -**gram LM**, that is

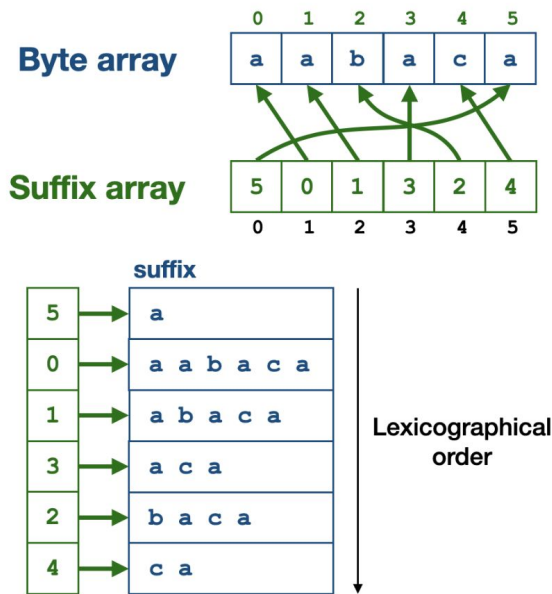
- trained on 5 trillion tokens
- contains n -gram counts for ~5 quadrillion (a.k.a. 5 thousand trillion) unique n -grams.

This n -gram LM is made possible by **backoff** and a special data structure. It can:

- compute n -gram LM probabilities and next-token distributions (like classic n -gram LMs)
- search for docs that contain specific n -grams
 - We can trace model responses back to snippets of training data!

So who even uses n -gram LMs today?

Infini-gram ([Liu et al. 2024](#)) is implemented using a *suffix array*



Train data

Hello, world!

Tokenized

Hello	,	world	!
-------	---	-------	---

Token IDs

0x3AC3	0x74C4	0x0C72	0x7527
--------	--------	--------	--------

Byte array

(Little-endian)

0x00	0x02	0x04	0x06
C3	3A	C4	74
72	0C	27	75

Suffix array

0x06	0x04	0x00	0x02
0x00	0x01	0x02	0x03

Downstream uses of infini-gram

Model: OLMo 2 13B Instruct

 Who is Celine Dion?

 Celine Dion is a Canadian singer known for her powerful voice and wide vocal range. She has achieved international fame and is one of the best-selling artists of all time. Born on March 30, 1968, in Charlemagne, Quebec, Canada, Dion began performing at a young age and gained prominence in the music industry in the 1980s.

Her career took off when she won the 1988 Eurovision Song Contest with the song "Ne partez pas sans moi." This victory launched her into international stardom. Dion is particularly known for her work in pop music, as well as her forays into genres such as rock, R&B, gospel, and classical.

Some of her most famous songs include "The Power of Love," "Beauty and the Beast" (for which she won an Academy Award for Best Original Song), "My Heart Will Go On" from the movie "Titanic," and "Because You Loved Me." She has sold over 200 million records worldwide and has won numerous awards, including five Grammy Awards, three American Music Awards, and several Billboard Music Awards.

In addition to her music career, Dion has also ventured into show business, including a long-running Las Vegas residency at the Colosseum at Caesars Palace, and she has been involved in philanthropy and humanitarian efforts. She is married to René Angélil, her former manager, and they have a son named René-Charles.

     Hide OLMoTrace

OLMoTrace

Documents from the training data that have exact text matches with the model response.
Powered by [infini-gram](#)
[More about how OLMoTrace works](#)

HIGH RELEVANCE

Pre-training document from:
[olmo-mix-1124](#) > web corpus (DCLM)

"...singer Celine Dion dominated the pop charts in the 1990s. She remains a popular entertainer and is the second-highest earning musician in history. **Born on March 30, 1968, in Charlemagne, Quebec, Canada,** singer Celine Dion had recorded nine French albums and won numerous awards by the time she was 18. She recorded her first English language album, Unison, in 1990..."

[View Document](#) [Locate span](#)

Post-training (SFT) document from:
[tulu-3-sft-olmo-2-mixture](#)

"...music styles. Born in Charlemagne, Quebec, she grew up in a family where music was a way of life, and she began her music career at a young age. Her music breakthrough came **when she won the 1988 Eurovision Song Contest,** which showcased her music talent to a global

OLMoTrace

Model: OLMo Documents from the training data that have exact text matches with the model

Pre-training document from: [olmo-mix-1124](#) 

> web corpus (DCLM)

URL: <http://easysector.com/celine-dion-tour-dates/>

"...Celine Dion Tickets

Celine Dion Tour Dates 2017

Celine Dion

French-Canadian singer Celine Dion dominated the pop charts in the 1990s. She remains a popular entertainer and is the second-highest earning musician in history. **Born on March 30, 1968, in Charlemagne, Quebec, Canada,** singer Celine Dion had recorded nine French albums and won numerous awards by the time she was 18. She recorded her first English language album, Unison, in 1990. Dion's real breakthrough into pop music stardom came in 1992, when she recorded the theme to Disney's hit animated feature Beauty and the Beast. She went on to record several hits, including four No. 1s: "The Power of Love," "Because You Loved Me," "My Heart Will Go On" and "I'm Your Angel." She has an angelic voice that many argue is the best in the business, and her astounding 200 million-plus albums sold worldwide is a testament to her talent. Now you can buy Celine Dion tickets and see why this music legend consistently tops the charts and delights audiences with each breathtaking show..."

[has been in](#) [won the 1988 Eurovision Song Contest,](#) which showcased her music

Problems with classic n -gram language models

Sparsity issues

What if the phrase in the numerator never occurs in the data?

The probability of valid sequences at test time would become 0!

$$P(x_i \mid x_{i-1}, \dots, x_{i-n+1}) \approx \frac{\text{count}(x_{i-n+1} \dots x_{i-2} \textcolor{teal}{x_{i-1}} \textcolor{teal}{x_i})}{\text{count}(x_{i-n+1} \dots x_{i-2} x_{i-1})}$$

Sparsity gets worse with higher n

Problems with classic n -gram language models

Sparsity issues

What if the phrase in the numerator never occurs in the data?

Smoothing: Add a small amount to the count for every word.

This is a type of *regularization*.

$$P(x_i \mid x_{i-1}, \dots, x_{i-n+1}) \approx \frac{\text{count}(x_{i-n+1} \dots x_{i-2} x_{i-1} x_i) + \delta}{\text{count}(x_{i-n+1} \dots x_{i-2} x_{i-1}) + |V|\delta}$$

Problems with classic n -gram language models

Sparsity issues

What if the denominator never occurs in the data?

Then we divide by zero!

$$P(x_i \mid x_{i-1}, \dots, x_{i-n+1}) \approx \frac{\text{count}(x_{i-n+1} \dots x_{i-2} \ x_{i-1} \ x_i)}{\text{count}(\textcolor{teal}{x}_{i-n+1} \dots \textcolor{teal}{x}_{i-2} \ \textcolor{teal}{x}_{i-1})}$$

Problems with classic n -gram language models

Sparsity issues

What if the denominator never occurs in the data?

Backoff: condition on $x_{i-1}, \dots, x_{i-n+2}$ instead

$$P(x_i \mid x_{i-1}, \dots, x_{i-n+1}) \approx P(x_i \mid x_{i-1}, \dots, x_{i-n+2})$$

Need to discount the higher-order n -gram to prevent $\text{sum}(\text{prob}) > 1$

Problems with classic n -gram language models

Sparsity issues

Interpolation: mix unigram, bigram, trigram, etc

$$\hat{P}(x_i | x_{i-1}, x_{i-2}) = \lambda_1 P(x_i | x_{i-1}, x_{i-2}) \\ + \lambda_2 P(x_i | x_{i-1}) + \lambda_3 P(x_i)$$

$$\sum_i \lambda_i = 1$$

Choose λ s to maximize probability of held-out data:
1) Calculate n -gram probs on training data
2) Search for λ s that give largest prob for heldout set

Problems with classic n -gram language models

No notion of semantic similarity

She bought a car

She purchased a bicycle

Cannot handle long-distance dependencies

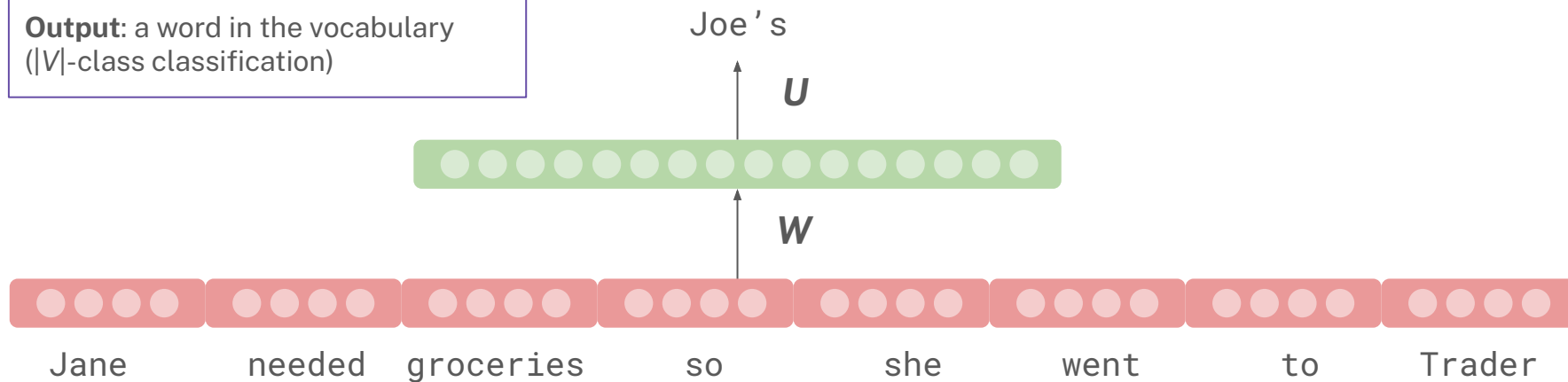
For tennis class he wanted to buy his own racquet

For programming class he wanted to buy his own computer

Fixed window neural LMs

A window-based neural model turns word prediction into **discriminative text classification**

Output: a word in the vocabulary
($|V|$ -class classification)



Input: $n-1$ previous words as context

Fixed window neural LMs

We replace the count table with a neural network to approximate the next-word prediction probability

θ are the weights (e.g. W, U) of our model

$$P_{\theta}(x_i | x_{i-1}, \dots, x_{i-n+1}) = \text{Softmax}(\text{NeuralNet}(x_{i-n+1}, \dots, x_{i-1}))$$

We can use SGD to optimize the network by minimizing the **average negative log-likelihood** of next word prediction

We sample pairs of (prefix context, next word) from a large dataset

$$\mathcal{L}(\theta) = -\mathbb{E}_{x_i | x_{<i}} \log P_{\theta}(x_i | x_{i-1}, \dots, x_{i-n+1})$$

Fixed window neural LMs

Produce an **output distribution**

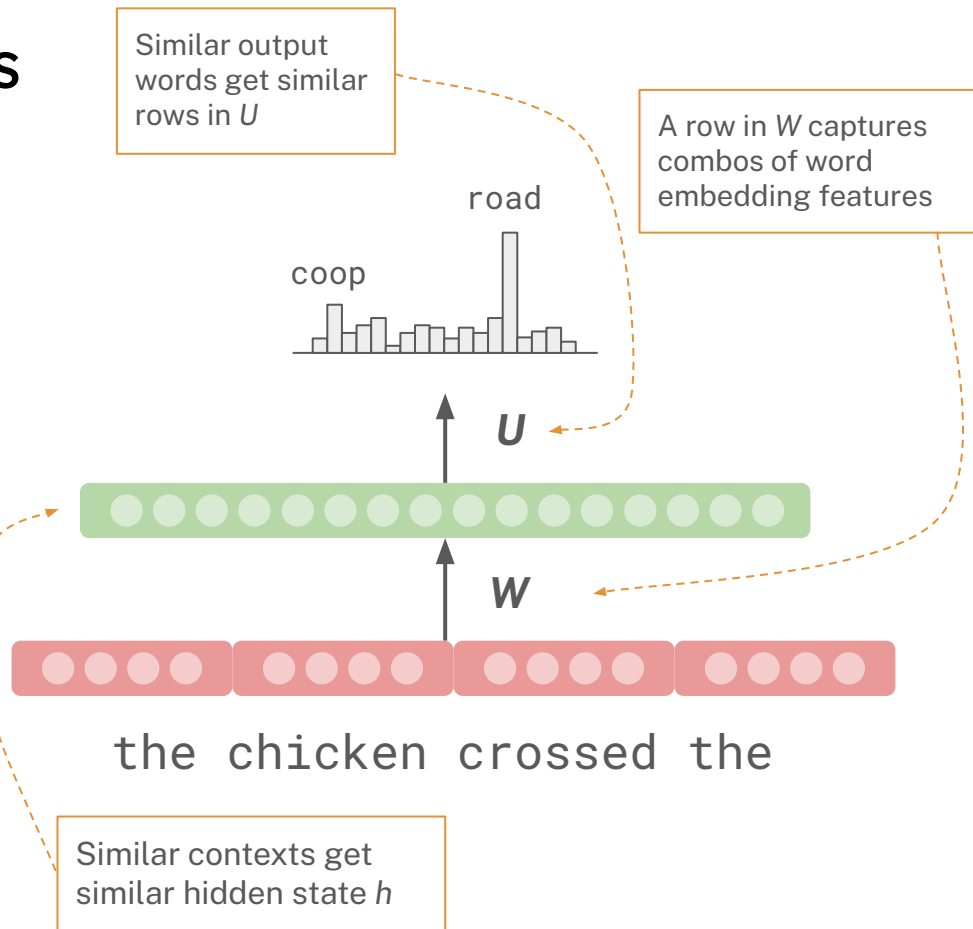
$$\hat{y} = \text{softmax}(Uh + b_2) \in \mathbb{R}^{|V|}$$

Hidden layer

$$h = f(We + b_1)$$

Concatenated word embeddings

$$e = [e_1; e_2; e_3; e_4]$$

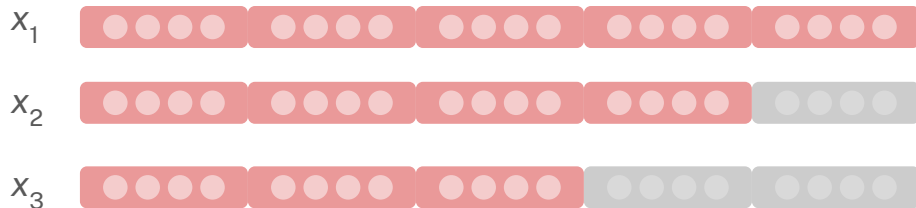


What if we have variable-sized inputs?

Let's say we have a window of size 5, but some of our input sequences are shorter than that.

x_1 = Alice went the store.	5 tokens
x_2 = Bob ate cheese.	4 tokens
x_3 = Joe laughed.	3 tokens

Solution: zero-pad up to the length of the longest sequence



Problems solved

- A neural LM learns to **count** *and* it learns to **compress**
 - Don't need to store all observed n -grams
 - Only need to store model parameters (weights)
 - No sparsity issues!
- Neural LMs can memorize texts and also **generalize**
 - Can share strength among similar words, and enforce implicit similarity within the model
 - Similar text inputs produce similar hidden representations
 - Similar text outputs come from similar hidden representations

Problems not solved

We still can't handle **long range dependencies**

For `tennis` class he wanted to get there early ... Later he stopped at the front desk to buy a new `racquet`

- A fixed window is **too small**
- Enlarging window enlarges W
- Each input embedding is multiplied by completely different weights in W ; there's no sharing or symmetry in how input words are processed

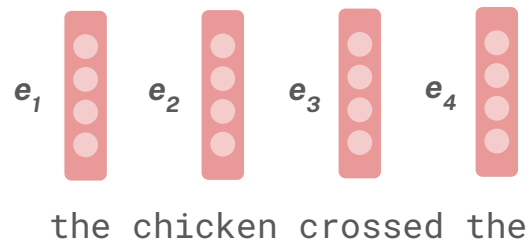
We need a neural architecture that can process *any length input*!

RNNs

Recurrent Neural Networks

Word embeddings

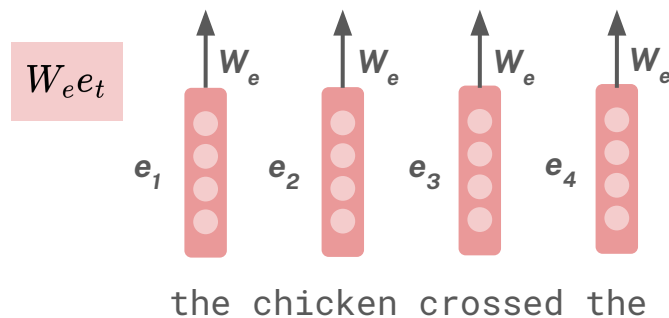
For an input sequence of any length



Recurrent Neural Networks

Word embeddings

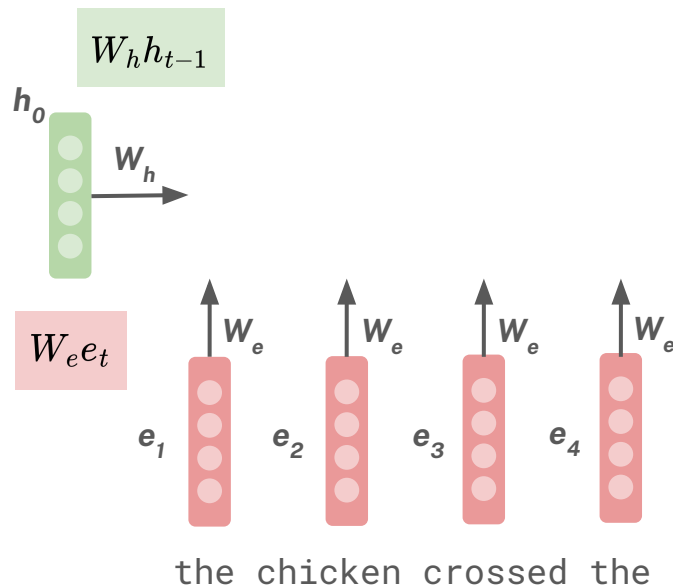
For an input sequence of any length



Recurrent Neural Networks

Word embeddings

For an input sequence of any length

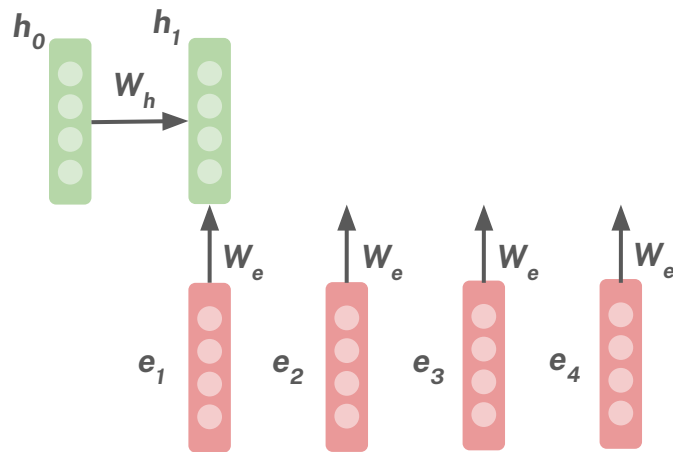


Recurrent Neural Networks

$$h_t = \sigma(W_h h_{t-1} + W_e e_t + b_1)$$

Word embeddings

For an input sequence of any length



Recurrent Neural Networks

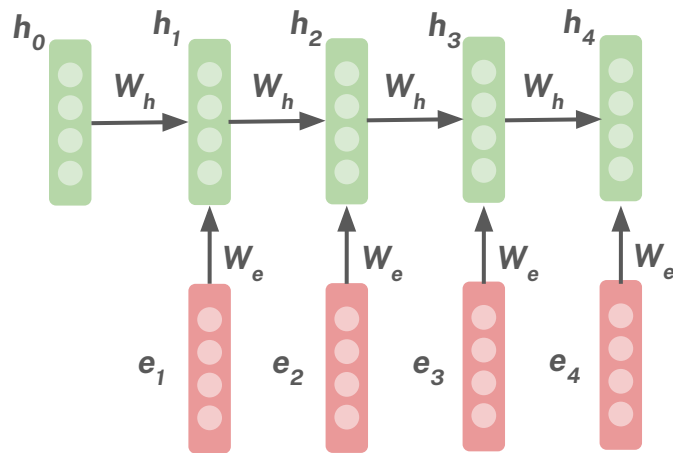
Hidden states

$$h_t = \sigma(W_h h_{t-1} + W_e e_t + b_1)$$

Key idea: Apply the same weights W repeatedly

Word embeddings

For an input sequence of any length



the chicken crossed the

Recurrent Neural Networks

Output probability distribution

$$\hat{y}_t = \text{softmax}(Uh_t + b_2) \in \mathbb{R}^{|V|}; \quad \hat{y}_4 = P(x_5 \mid \text{the chicken crossed the})$$

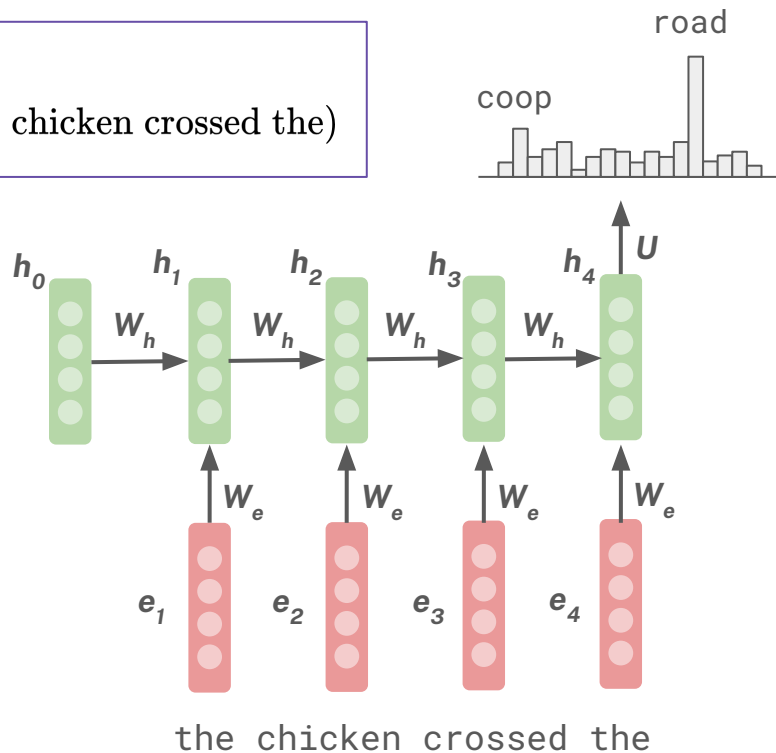
Hidden states

$$h_t = \sigma(W_h h_{t-1} + W_e e_t + b_1)$$

Key idea: Apply the same weights W repeatedly

Word embeddings

For an input sequence of any length



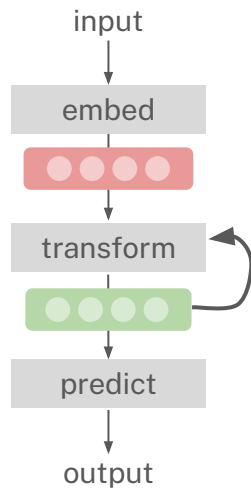
Another view of RNNs

“Remembers” information along the
“time dimension”

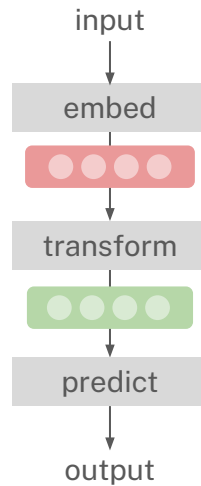
RNNs are neural nets that share
weights across layers, takes in an input
at each layer, w/ a variable number of
layers

The *transform* step gets its own
“previous” value as inputs

Recurrent NN



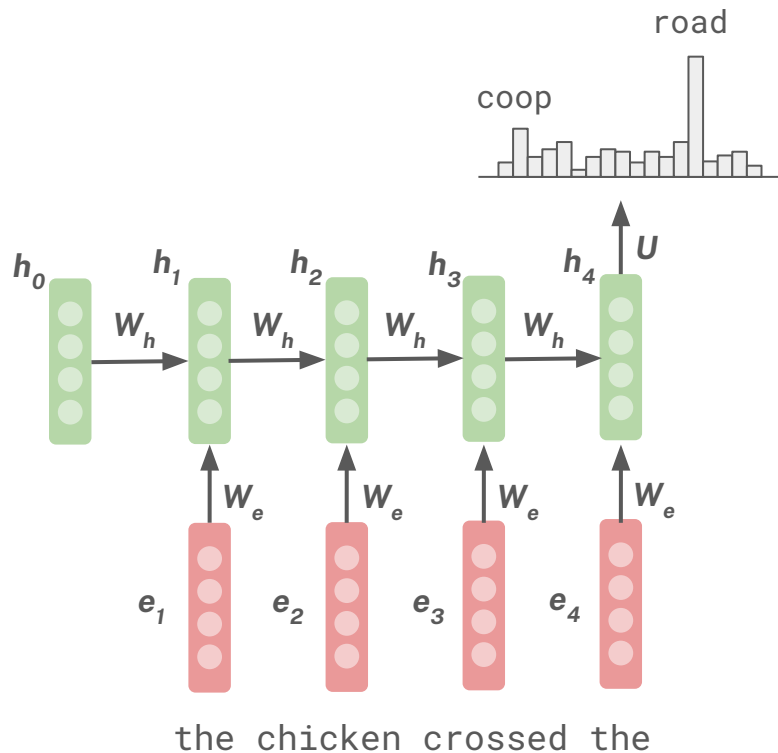
Feed-forward NN



Recurrent Neural Networks

Advantages ✓

- Flexible for use with any input length
- Computation for step t can use information from multiple steps back
- Model size doesn't increase for longer inputs
- Same weights applied on every timestep, so there is symmetry in how inputs are processed

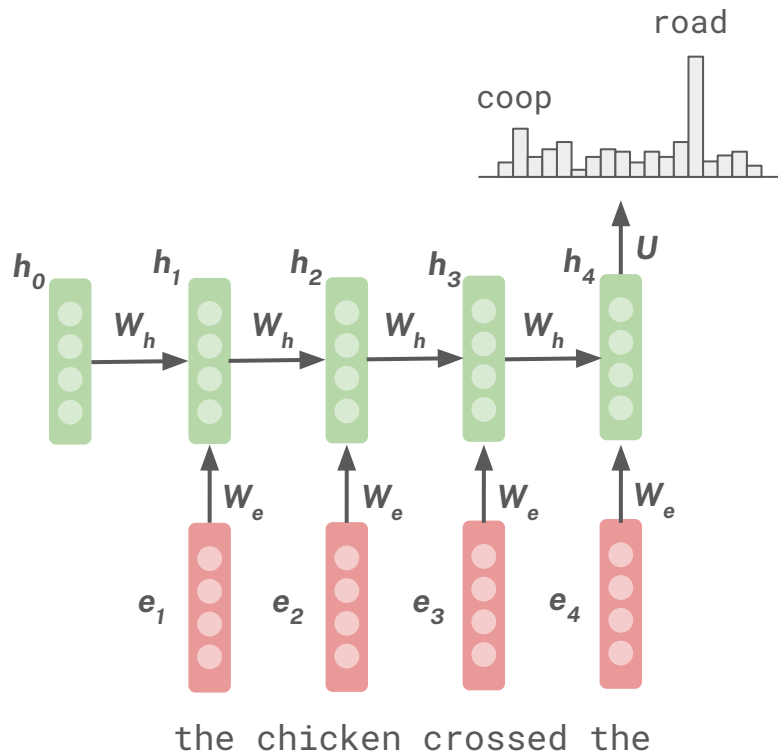


Recurrent Neural Networks

Disadvantages ✗

- Recurrent computation is slow
- In practice, difficult to access info from many steps back

More on these points later!



Training an RNN LM

Get a big dataset of text with a sequence of words $x_1, \dots, x_t, x_{t+1}, \dots, x_T$

Training an RNN LM

Get a big dataset of text with a sequence of words $x_1, \dots, x_t, x_{t+1}, \dots, x_T$

Feed into an RNN LM, and compute output distributions at every step t for every word

Training an RNN LM

Get a big dataset of text with a sequence of words $x_1, \dots, x_t, x_{t+1}, \dots, x_T$

Feed into an RNN LM, and compute output distributions at every step t for every word

At each step, calculate loss as the **cross-entropy** between the predicted probability distribution and the true next word x_{t+1} .

Training an RNN LM

Get a big dataset of text with a sequence of words $x_1, \dots, x_t, x_{t+1}, \dots, x_T$

Feed into an RNN LM, and compute output distributions at every step t for every word

At each step, calculate loss as the **cross-entropy** between the predicted probability distribution and the true next word x_{t+1} .

Let y_t be a one-hot vector representing the “true” distribution at that step and $\hat{y}_t$ is the predicted probability distribution:

$$J_t(\theta) = CE(y_t, \hat{y}_t)$$

Training an RNN LM

Get a big dataset of text with a sequence of words $x_1, \dots, x_t, x_{t+1}, \dots, x_T$

Feed into an RNN LM, and compute output distributions at every step t for every word

At each step, calculate loss as the **cross-entropy** between the predicted probability distribution and the true next word x_{t+1} .

Let y_t be a one-hot vector representing the “true” distribution at that step and $\hat{y}_t$ is the predicted probability distribution:

$$J_t(\theta) = CE(y_t, \hat{y}_t) = - \sum_{w \in V} y_t(w) \log \hat{y}_t(w)$$

Training an RNN LM

Get a big dataset of text with a sequence of words $x_1, \dots, x_t, x_{t+1}, \dots, x_T$

Feed into an RNN LM, and compute output distributions at every step t for every word

At each step, calculate loss as the **cross-entropy** between the predicted probability distribution and the true next word x_{t+1} .

Let y_t be a one-hot vector representing the “true” distribution at that step and $\hat{y}_t$ is the predicted probability distribution:

$$J_t(\theta) = CE(y_t, \hat{y}_t) = - \sum_{w \in V} y_t(w) \log \hat{y}_t(w)$$

$$y_t(w) = \begin{cases} 1 & w = x_{t+1} \\ 0 & w \neq x_{t+1} \end{cases}$$

Training an RNN LM

Get a big dataset of text with a sequence of words $x_1, \dots, x_t, x_{t+1}, \dots, x_T$

Feed into an RNN LM, and compute output distributions at every step t for every word

At each step, calculate loss as the **cross-entropy** between the predicted probability distribution and the true next word x_{t+1} .

Let y_t be a one-hot vector representing the “true” distribution at that step and $\hat{y}_t$ is the predicted probability distribution:

$$J_t(\theta) = CE(y_t, \hat{y}_t) = - \sum_{w \in V} y_t(w) \log \hat{y}_t(w) = - \log \hat{y}_t(x_{t+1}) \quad y_t(w) = \begin{cases} 1 & w = x_{t+1} \\ 0 & w \neq x_{t+1} \end{cases}$$

Training an RNN LM

Get a big dataset of text with a sequence of words $x_1, \dots, x_t, x_{t+1}, \dots, x_T$

Feed into an RNN LM, and compute output distributions at every step t for every word

At each step, calculate loss as the **cross-entropy** between the predicted probability distribution and the true next word x_{t+1} .

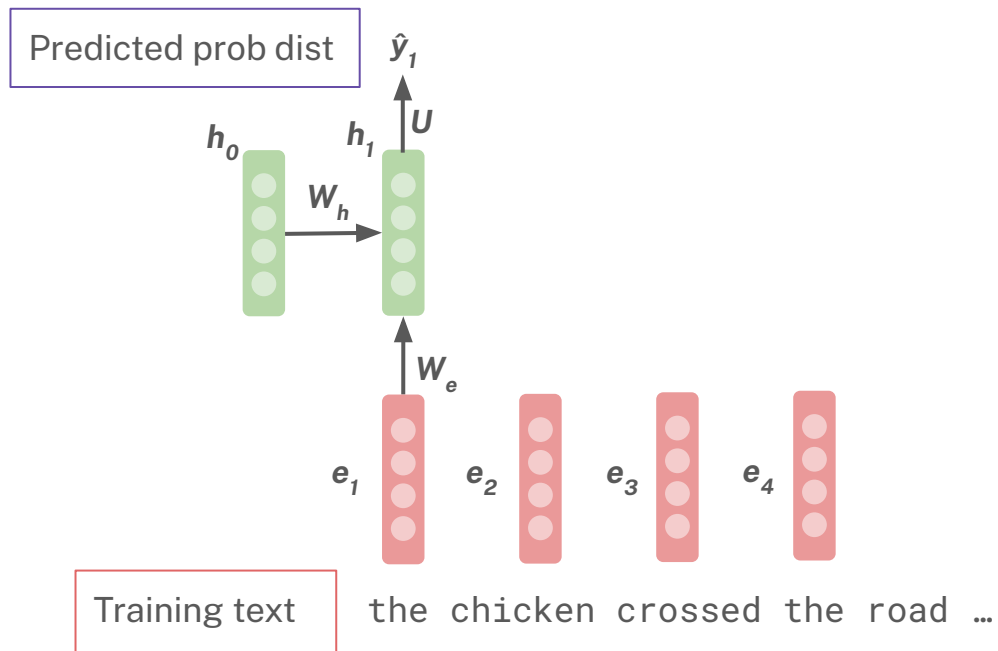
Let y_t be a one-hot vector representing the “true” distribution at that step and $\hat{y}_t$ is the predicted probability distribution:

$$J_t(\theta) = CE(y_t, \hat{y}_t) = - \sum_{w \in V} y_t(w) \log \hat{y}_t(w) = - \log \hat{y}_t(x_{t+1}) \quad y_t(w) = \begin{cases} 1 & w = x_{t+1} \\ 0 & w \neq x_{t+1} \end{cases}$$

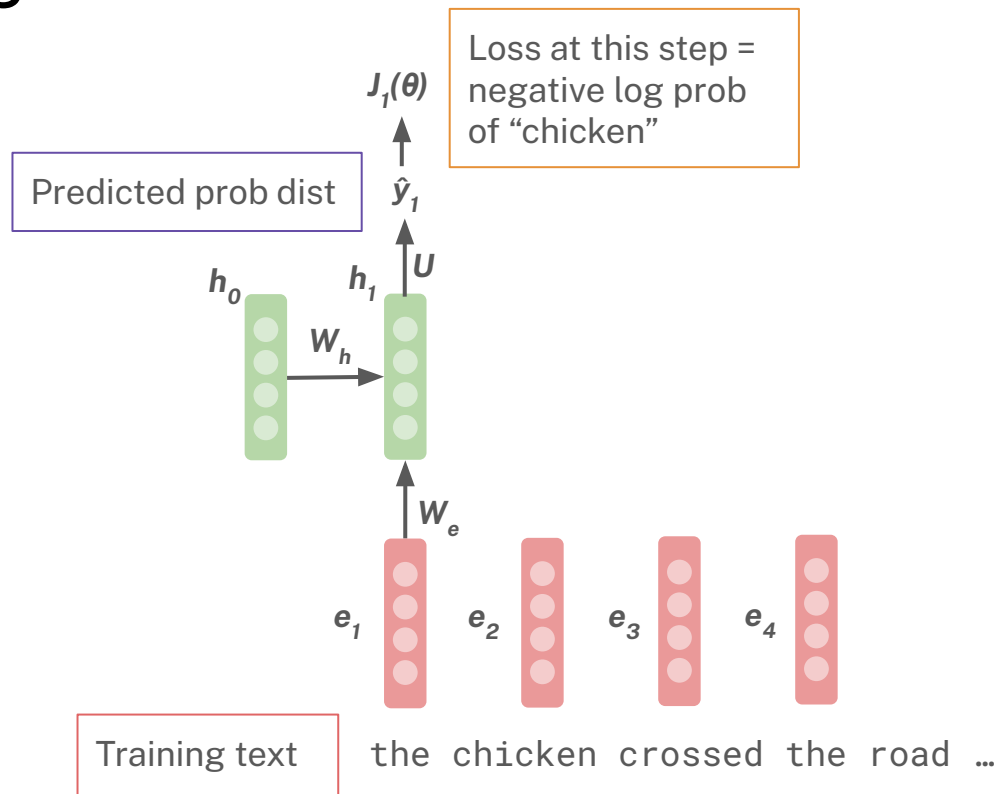
Average these losses to get overall loss for entire training set:

$$J(\theta) = \frac{1}{T} \sum_{t=1}^T J_t(\theta) = \frac{1}{T} \sum_{t=1}^T - \log \hat{y}_t(x_{t+1})$$

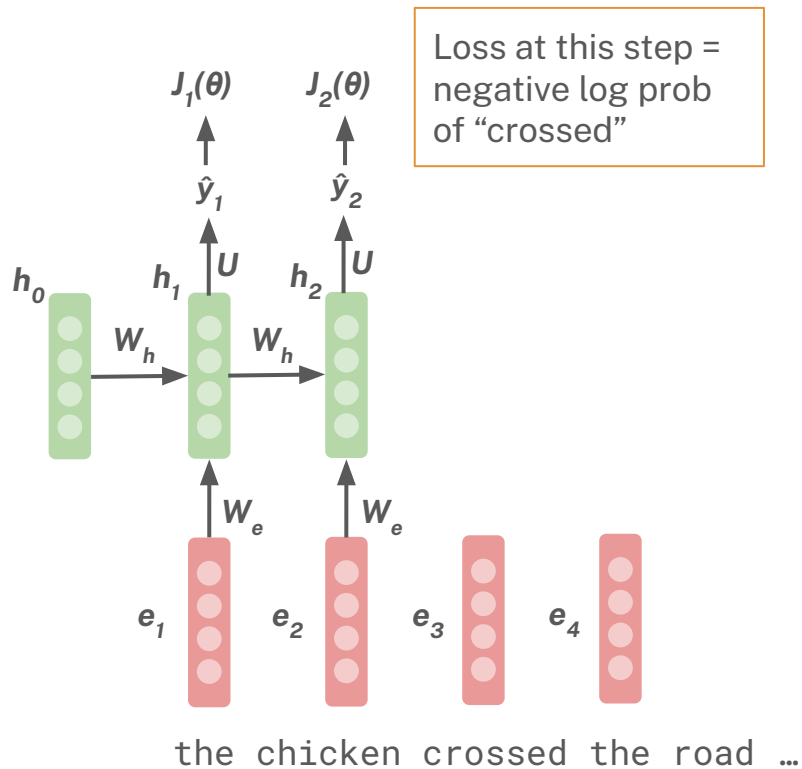
Training an RNN LM



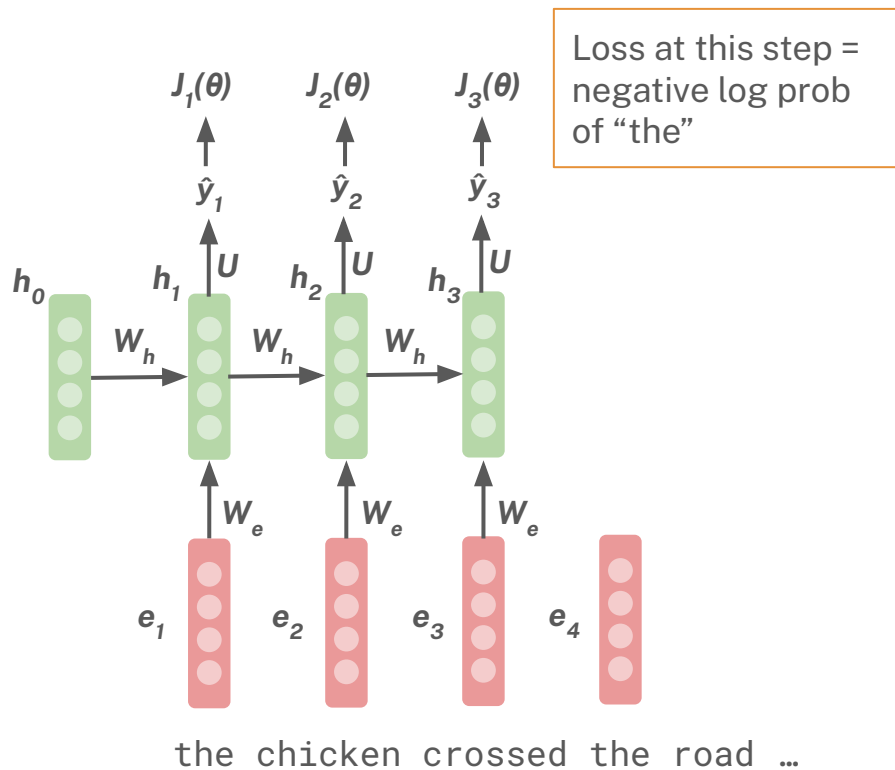
Training an RNN LM



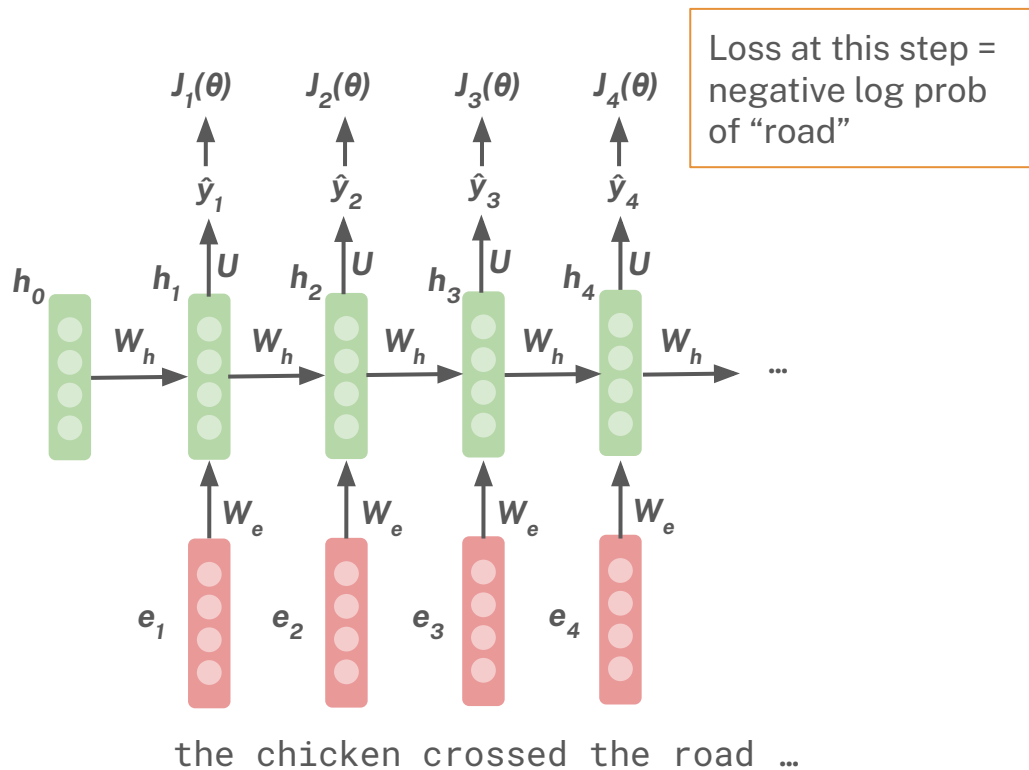
Training an RNN LM



Training an RNN LM

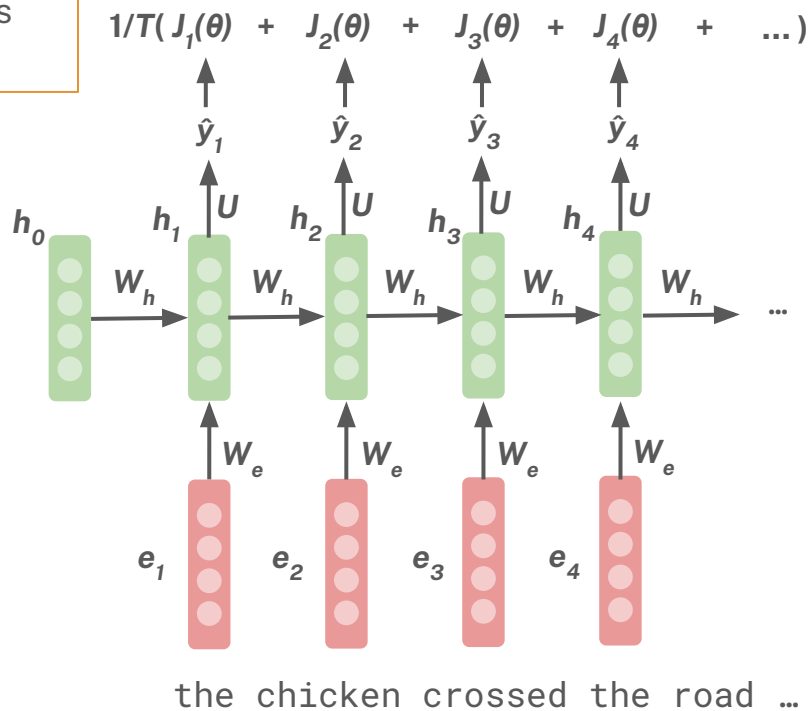


Training an RNN LM



Training an RNN LM

Average the loss across all steps



Teacher forcing is a training technique where ground-truth from the training dataset is fed as the input for each time step, instead of the model's own prediction.

Training an RNN LM

Computing loss and gradients across an entire dataset at once is expensive, memory-wise.

Instead: compute loss for a **batch of sentences**, and then compute gradients, update our weights, and repeat.

Selecting the right batch size (when training any type of LM) is a trade-off b/t hardware memory, training speed, and model accuracy.

- **Too small:** noisy updates could help escape local minima, but trains slower, might underutilize compute
- **Too large:** fast & smooth updates, but memory hungry and may cause you to converge to suboptimal minima

Usually batch sizes are powers of two. Adjustments to batch size usually requires adjusting learning rate, too.

Backpropagation

Model needs to learn from its errors

→ We need to update our weights based on our loss, using SGD

So, how do we compute the gradient of our loss function?

Multivariable chain rule

To calculate the derivative of a function composed of two intermediate functions:

$$\frac{d}{dt} f(x(t), y(t)) = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}$$

When a variable influences the output through several paths, the total derivative is the sum of the derivatives along each path.

In an RNN, to update $\mathbf{W}_h$, we need to compute $\frac{\partial J}{\partial \mathbf{W}_h}$.

$\mathbf{W}_h$ is shared across “time”, influencing the output via multiple paths, at every time step. It appears in the intermediate computation of each hidden state:

$$h_t = \sigma(W_h h_{t-1} + W_e e_t + b_1)$$

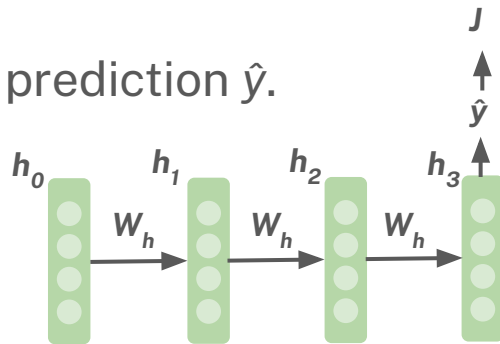
Applying the multivariable chain rule

The rule tells us: computing $\frac{\partial J}{\partial W_h}$ means we should sum up the derivatives from each computation W_h is involved in.

Example: let's say $t = 3$, where h_3 is used to make our prediction $\hat{y}$.

W_h influences J via three paths:

- Used to compute $h_3 \rightarrow$ affects J
- Used to compute $h_2 \rightarrow$ affects $h_3 \rightarrow$ affects J
- Used to compute $h_1 \rightarrow$ affects $h_2 \rightarrow$ affects $h_3 \rightarrow$ affects J



Applying the multivariable chain rule

Example: let's say $t = 3$, where h_3 is used to make our prediction $\hat{y}$.

W_h influences J via three paths, which get summed up based on the rule:

$$\frac{\partial J_3}{\partial W_h} = \frac{\partial J_3}{\partial h_3} \frac{\partial h_3}{\partial W_h} + \frac{\partial J_3}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial W_h} + \frac{\partial J_3}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial h_1} \frac{\partial h_1}{\partial W_h}$$

General formula:

$$\sum_{k=1}^t \frac{\partial J}{\partial h_t} \left(\prod_{j=k+1}^t \frac{\partial h_j}{\partial h_{j-1}} \right) \frac{\partial h_k}{\partial W_h}$$

In practice, backprop is “truncated” after
~20 timesteps for training efficiency

Applying the **chain rule** along this path: How h_3 changed the loss, times how h_2 changed h_3 , times how h_1 changed h_2 , times how W_h changed h_1 .

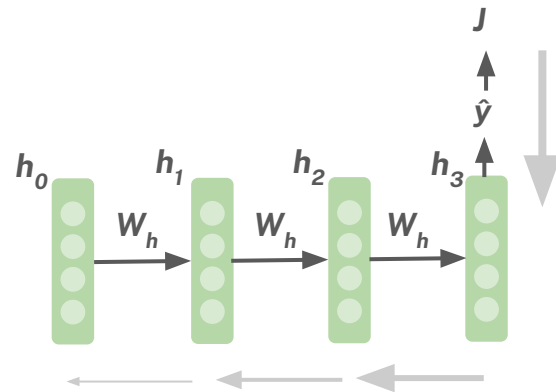
Vanishing gradients

What happens if the values in this product are small?

$$\sum_{k=1}^t \frac{\partial J}{\partial h_t} \left(\prod_{j=k+1}^t \frac{\partial h_j}{\partial h_{j-1}} \right) \frac{\partial h_k}{\partial W_h}$$

With non-linearities from RNNs' activation functions (e.g. tanh), it's easy to get derivatives less than 1.

The more small values we multiply, and the further we backprop, the gradient signal gets small and smaller!



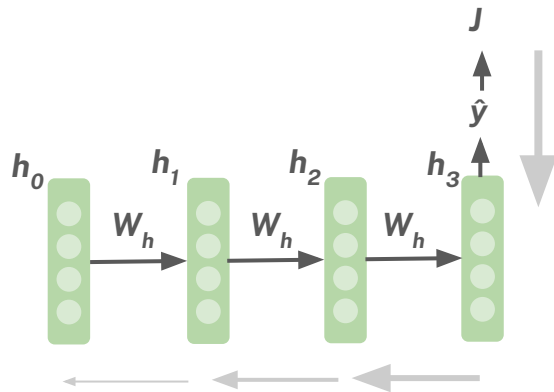
Vanishing gradients

What happens if the values in this product are small?

$$\sum_{k=1}^t \frac{\partial J}{\partial h_t} \left(\prod_{j=k+1}^t \frac{\partial h_j}{\partial h_{j-1}} \right) \frac{\partial h_k}{\partial W_h}$$

With non-linearities from RNNs' activation functions (e.g. tanh), it's easy to get derivatives less than 1.

This means model params are updated only with respect to near effects, rather than long-term effects!



Effect of vanishing gradient on RNN LM

When she tried to print her **tickets**, she found that the printer was out of toner. She went to the stationery store to buy more toner. It was very overpriced. After installing the toner into the printer, she finally printed her -----

To learn from this training example, the LM needs to model the dependency between *tickets* at the 7th step and *tickets* at the end.

But if the gradient is small, the LM can't learn this dependency

→ So it's unable to predict dependencies like this at test time.

How to fix vanishing gradients

It's difficult for the RNN to learn to preserve information over many timesteps.

This is because the hidden state is constantly being rewritten.

Example fix: an RNN architecture with a “memory” component

→ Long short-term memory (LSTM) [[link](#)]

Another fix: create more direct and linear pass-through connections

→ Residual connections, **attention**, etc

Exploding gradients

If the gradient becomes too big, then the SGD update step becomes too big:

$$\theta_{new} = \theta_{old} - \alpha \nabla_{\theta} J(\theta)$$

learning rate

gradient

This can cause bad updates: we take too large a step and reach a weird and bad parameter configuration, w/ a large loss.

You think you've found a hill to climb, but suddenly you're in Iowa!

In the worse case, this will result in *inf* or *nan* in your network and then you have to restart training from an earlier checkpoint.

How to fix exploding gradients

Gradient clipping: if the norm of the gradient is greater than some threshold, scale it down before applying SGD update

Algorithm 1 Pseudo-code for norm clipping the gradients whenever they explode

$$\hat{\mathbf{g}} \leftarrow \frac{\partial \mathcal{E}}{\partial \theta}$$
$$\text{if } \|\hat{\mathbf{g}}\| \geq \textit{threshold} \text{ then}$$
$$\quad \hat{\mathbf{g}} \leftarrow \frac{\textit{threshold}}{\|\hat{\mathbf{g}}\|} \hat{\mathbf{g}}$$
$$\text{end if}$$

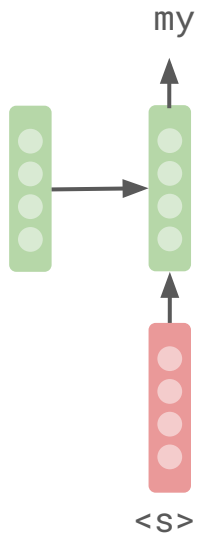
[Pascanu et al. 2012](#)

Take a step in the same direction, but a smaller step

Remembering to clip gradients is important, but exploding gradients have a simple solution!

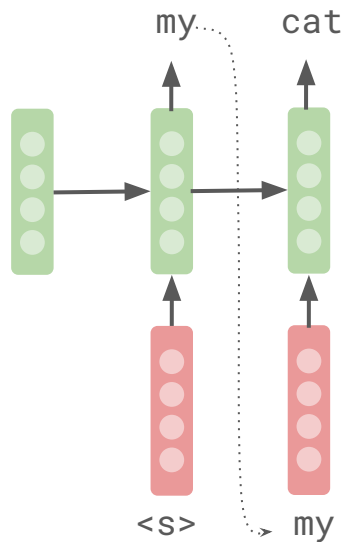
Generating roll outs

Just like with an n -gram language model, you can generate text by repeated sampling, in an **autoregressive** way. Sampled output = next step's input.



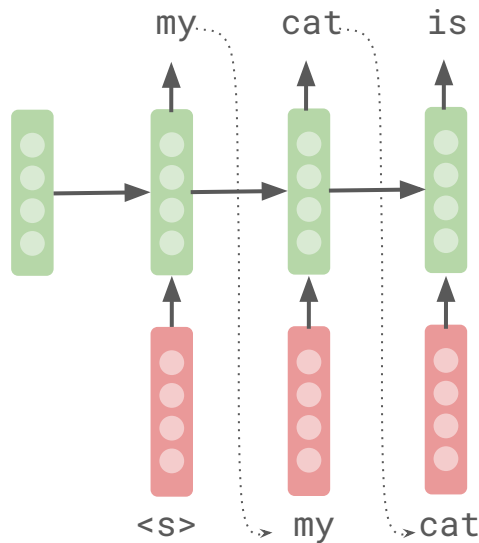
Generating roll outs

Just like with an n -gram language model, you can generate text by repeated sampling, in an **autoregressive** way. Sampled output = next step's input.



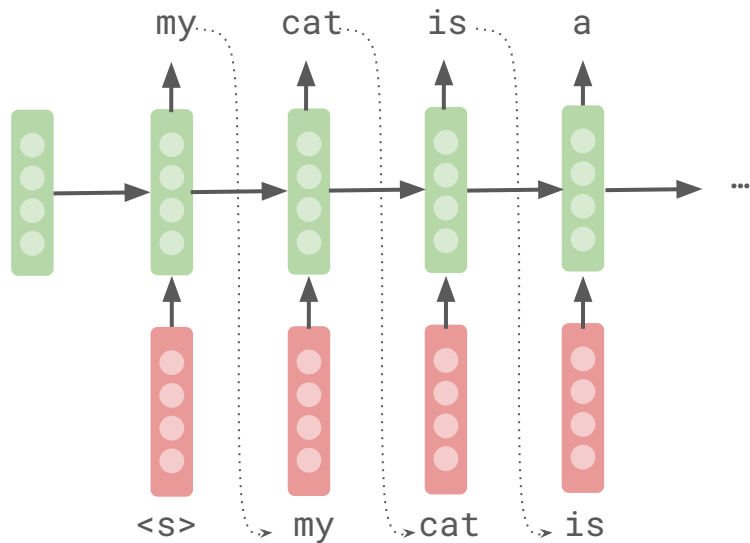
Generating roll outs

Just like with an n -gram language model, you can generate text by repeated sampling, in an **autoregressive** way. Sampled output = next step's input.



Generating roll outs

Just like with an n -gram language model, you can generate text by repeated sampling, in an **autoregressive** way. Sampled output = next step's input.



Generating roll outs

Train an RNN on recipes, and then generate ([source](#)):

Title: CHOCOLATE RANCH BARBECUE

Categories: Game, Casseroles, Cookies, Cookies

Yield: 6 Servings

2 tb Parmesan cheese -- chopped

1 c Coconut milk

3 Eggs, beaten

Place each pasta over layers of lumps. Shape mixture into the moderate oven and simmer until firm. Serve hot in bodied fresh, mustard, orange and cheese.

Combine the cheese and salt together the dough in a large skillet; add the ingredients and stir in the chocolate and pepper.

Evaluation

The standard evaluation metric for language models is **perplexity**.

$$\text{perplexity} = \prod_{t=1}^T \left(\frac{1}{P_{LM}(x_{t+1} | x_t, \dots, x_1)} \right)^{1/T}$$

Geometric mean, so it's normalized by the # of words

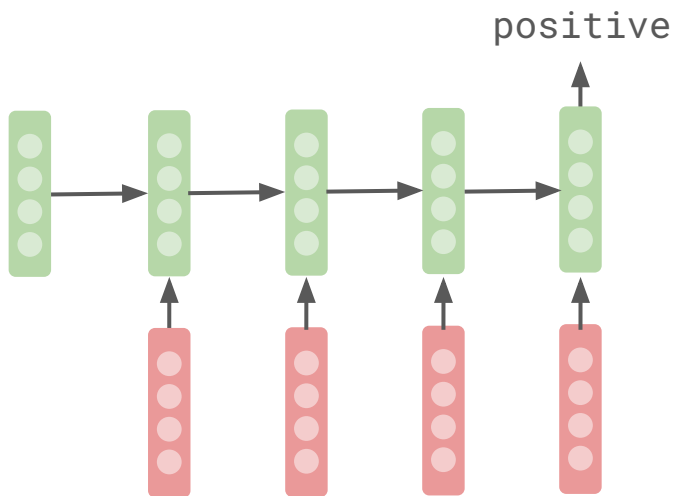
Inverse probability of dataset, according to the language model

$$= \prod_{t=1}^T \left(\frac{1}{\hat{y}_t(x_{t+1})} \right)^{1/T} = \exp \left(\frac{1}{T} \sum_{t=1}^T -\log \hat{y}_t(x_{t+1}) \right) = \exp(J(\theta))$$

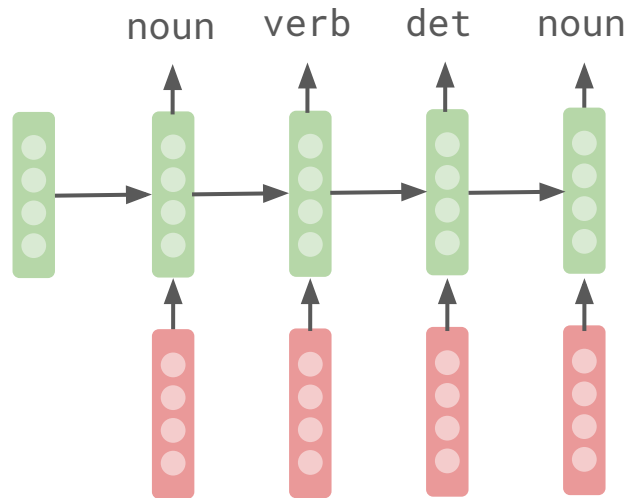
Exponential of the cross-entropy loss!

Lower perplexity is better.

Can use RNNs for other tasks too!



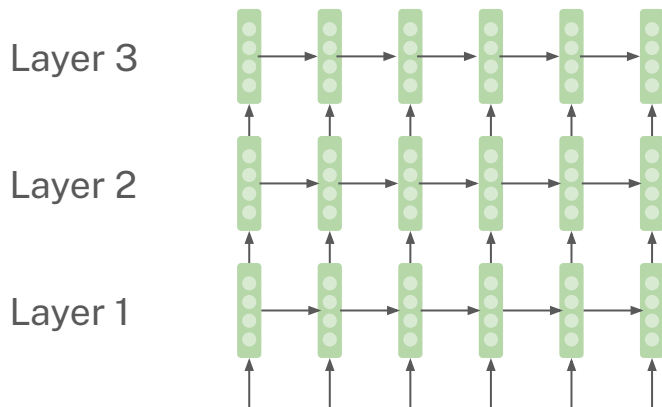
Encoding sentences for sentence classification. Can also use hidden states to get sentence embeddings!



Tagging sentences with labels for each word, e.g. part-of-speech, named entity recognition.

Back in my day, we RNN'd in all sorts of ways

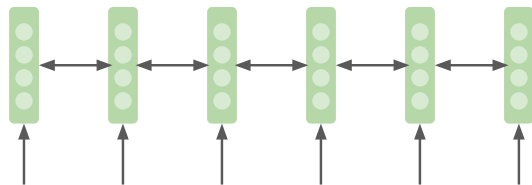
Multilayer RNNs: hidden states from layer i are inputs to layer $i + 1$.



Transformers (later lecture) can be up to 24 layers

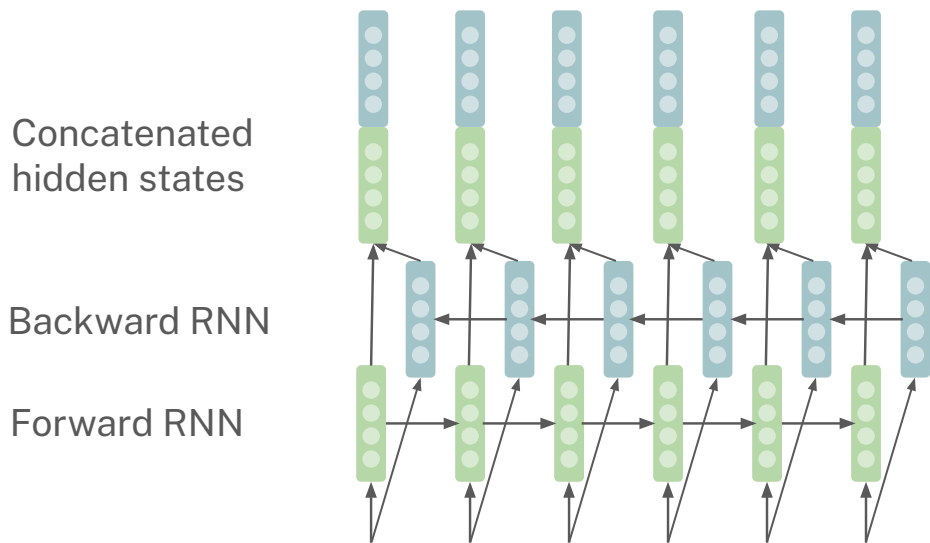
Back in my day, we RNN'd in all sorts of ways

Bidirectional RNNs



Back in my day, we RNN'd in all sorts of ways

Bidirectional RNNs



Question

Can we use bidirectional RNNs in the following tasks?

Text classification, sequence tagging, text generation

- (a) Yes, Yes, Yes
- (b) Yes, No, Yes
- (c) Yes, Yes, No
- (d) No, Yes, No

RNNs summary

Advantages

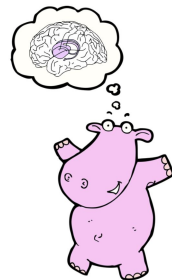
- Can process any length input
- Computation for each step can (in theory) use info from many steps back
- Model size doesn't increase for longer input context

Disadvantages

- Recurrent computation is slow and can't parallelize
- In practice, difficult to access information from many steps back

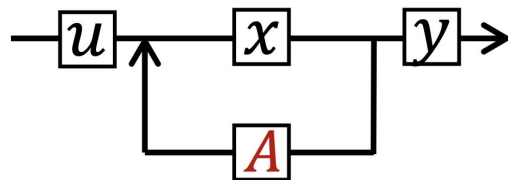
Many RNN variants!

We won't dive into a history lesson on LSTMs, GRUs, etc, but...



State-space models are modern RNNs

- Linear recurrent update
- Long-range with large hidden units
- Efficiency: parallelizable on GPUs
- [“A refreshing departure from Transformers”](#)
- [Mamba](#) is an example SSM



$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

From [the S4 paper](#)

Seq2seq Task Example: Machine Translation

Multilinguality

There are over 7,000 documented languages in the world!

You can browse languages of the world on [Ethnologue](#), from 'Auhelawa to Zuni.

To facilitate communication among people, *translation* is needed.

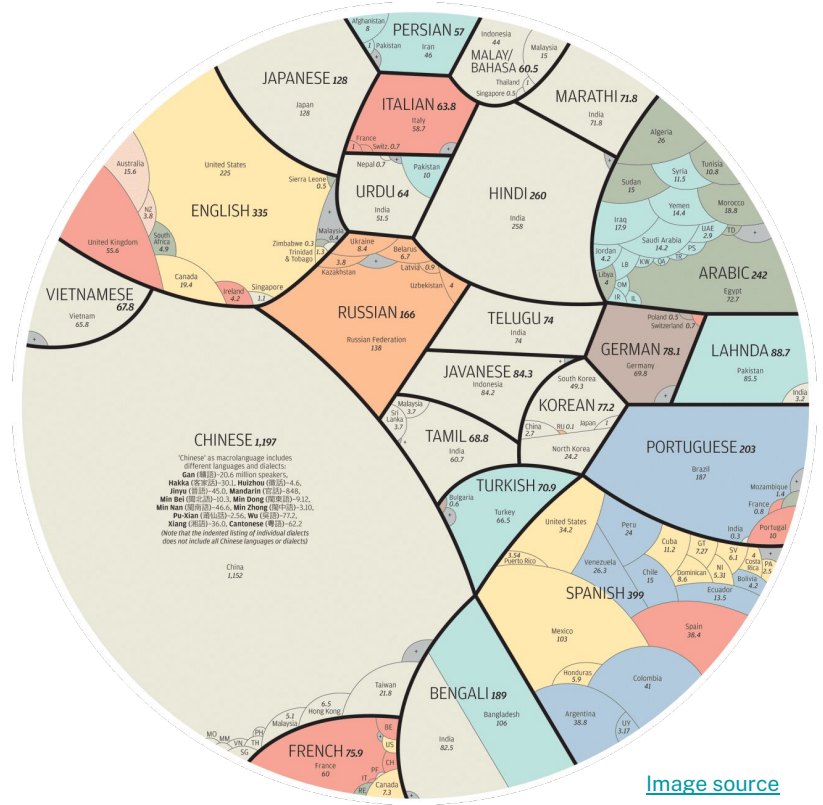


Image source

Machine translation

Goal: translate a sentence in a source language (input) to a sentence in a target language (output)

I like apples ↔ ich mag Äpfel (German)

For awhile, MT was one of the “holy grail” problems in AI

Still very challenging for low-resource languages

Machine translation

MT is challenging because

- Single words may be replaced with multi-word phrases:

I like **apples** ↔ J'aime **les pommes** (French)

- Phrases can be reordered

I like **red** apples ↔ J'aime les pommes **rouges** (French)

- Some translations are context-dependent

les ↔ **the** but **les** pommes ↔ apples

Extremely large output space! Decoding is NP-hard.

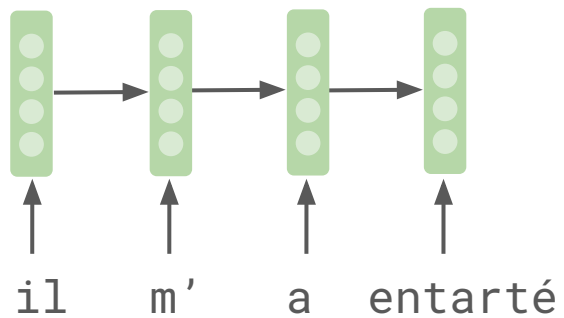
Neural machine translation

NMT was a **fringe research attempt** in 2014 and became a **leading standard method** in 2016.

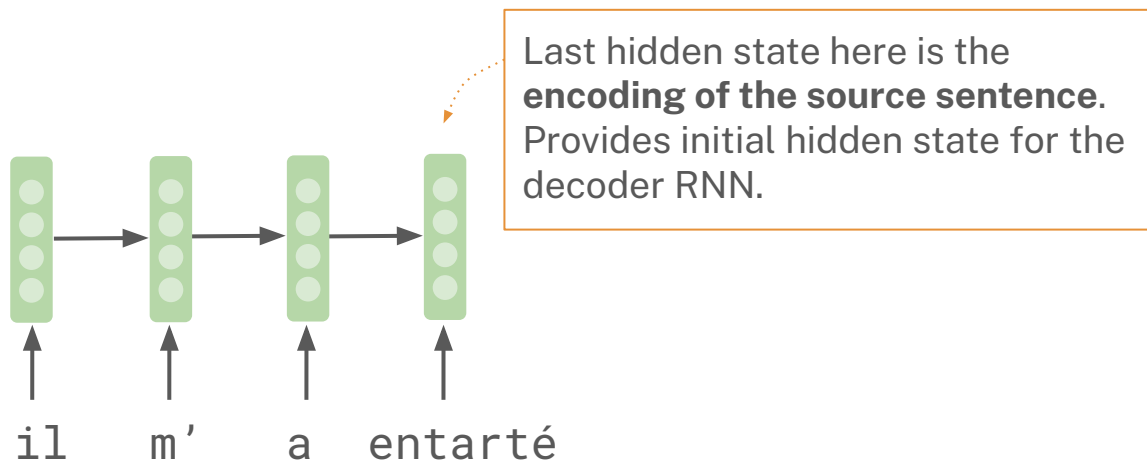
- 2014: first seq2seq paper published ([Sutskever et al. 2014](#))
- 2016: Google Translate switches from statistical machine translation (SMT) to NMT, and by 2018 everyone else (Microsoft, Facebook, Baidu, Tencent, etc) has as well

SMT systems, build by **hundreds of engineers over many years**, were outperformed by NMT systems trained by **small groups of engineers in a few months**.

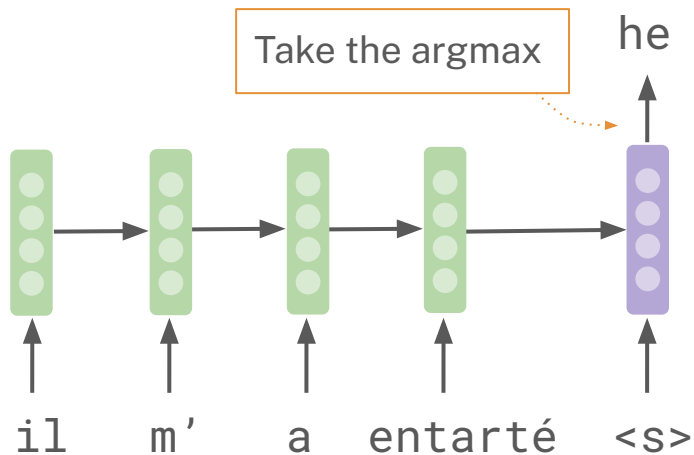
Neural machine translation



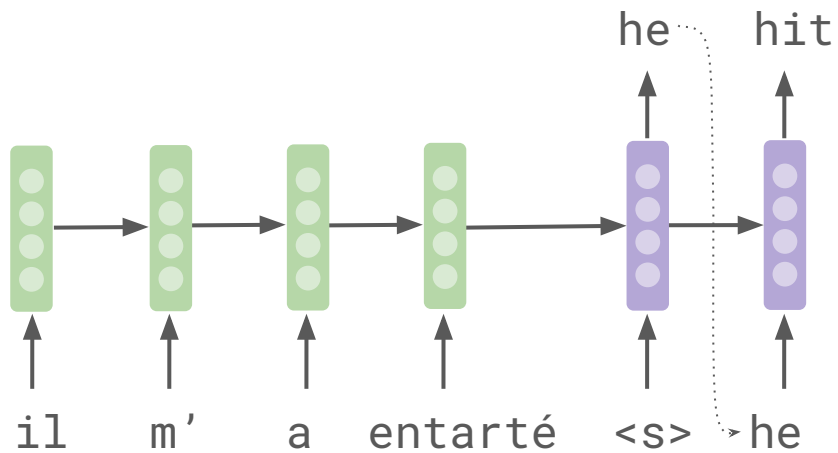
Neural machine translation



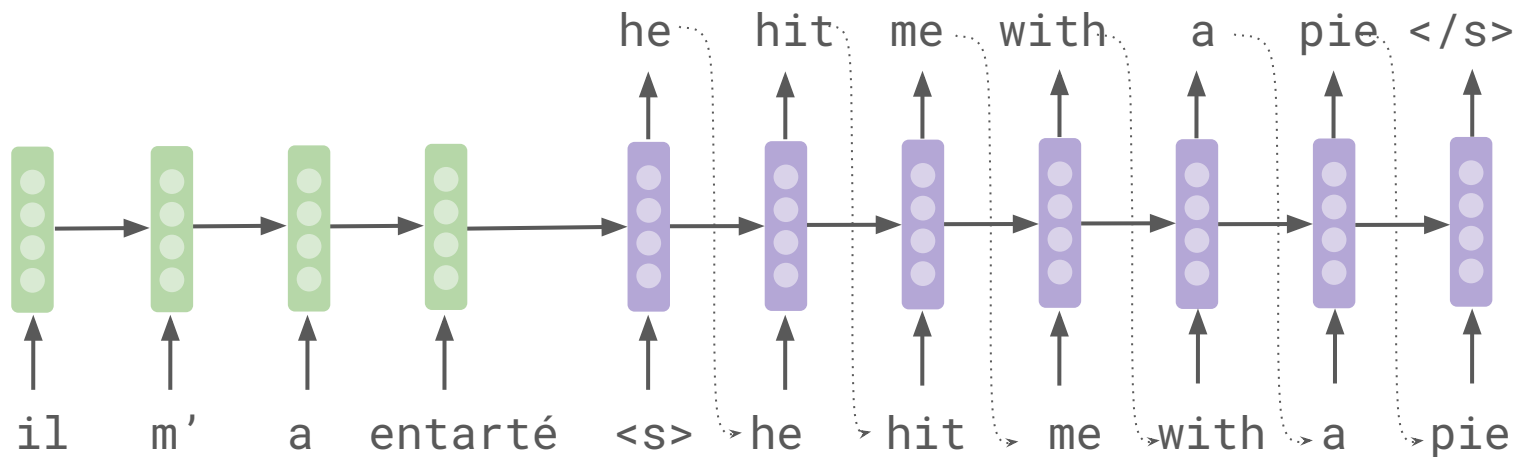
Neural machine translation



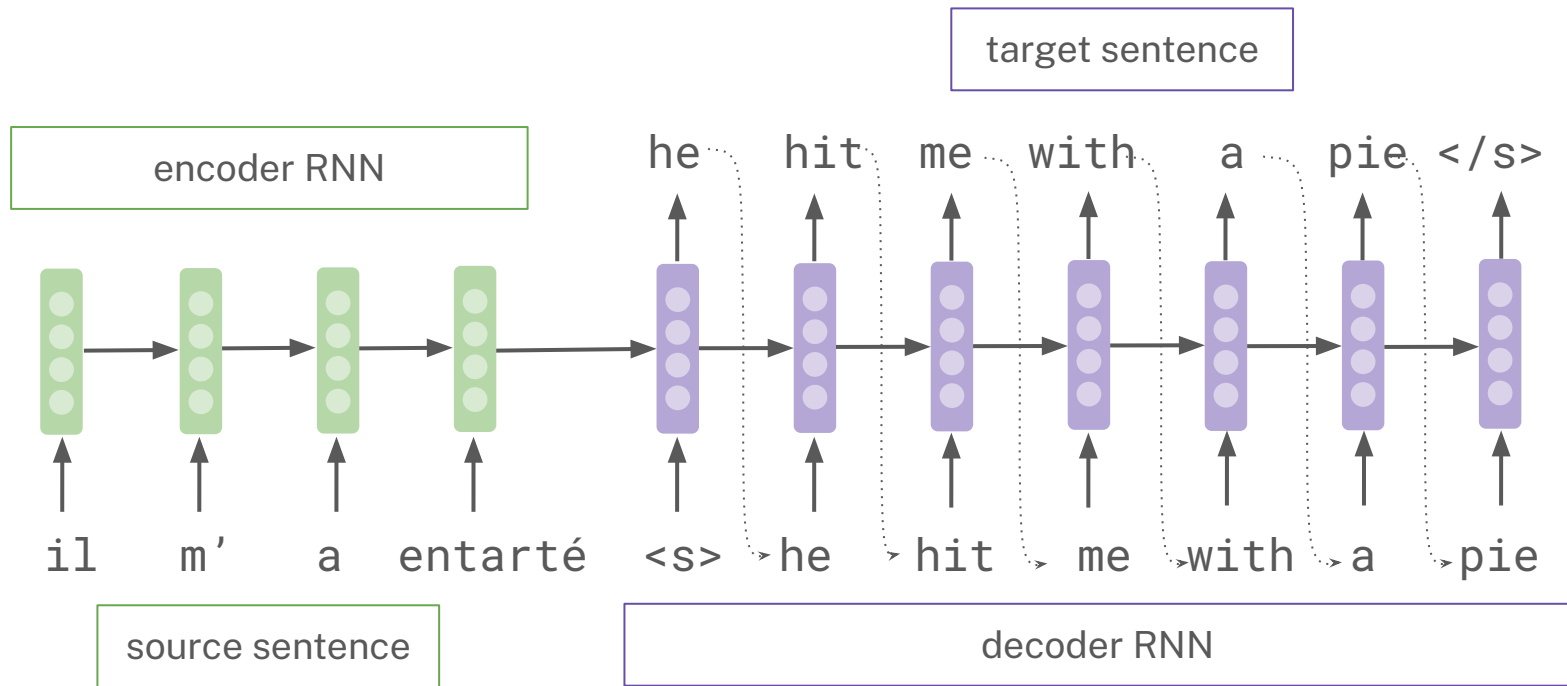
Neural machine translation



Neural machine translation

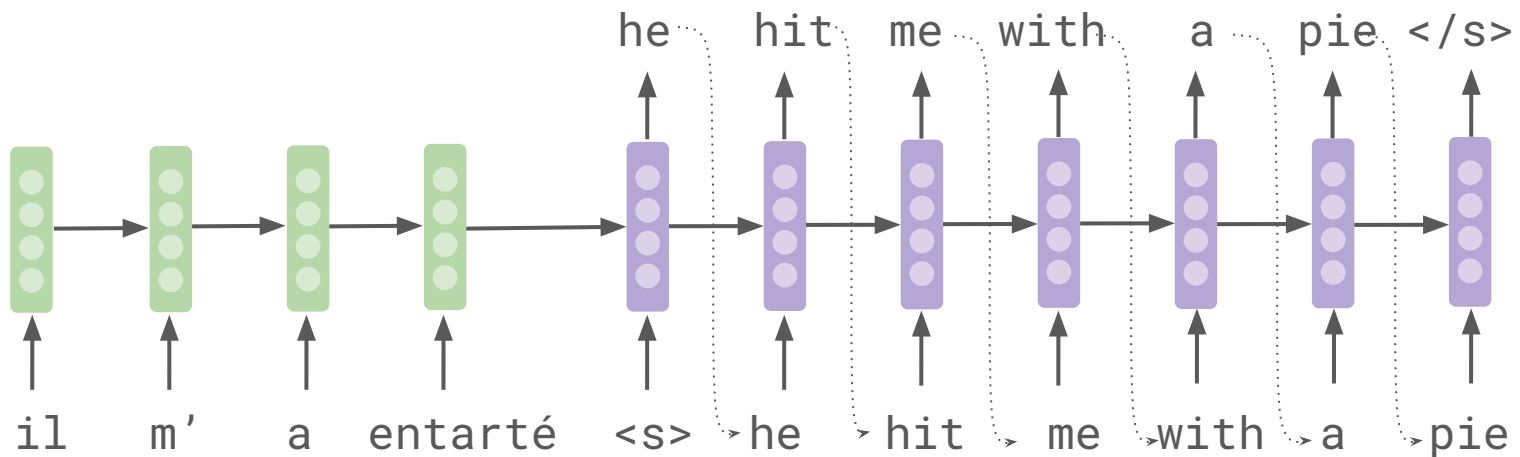


Neural machine translation



A sequence-to-sequence model!

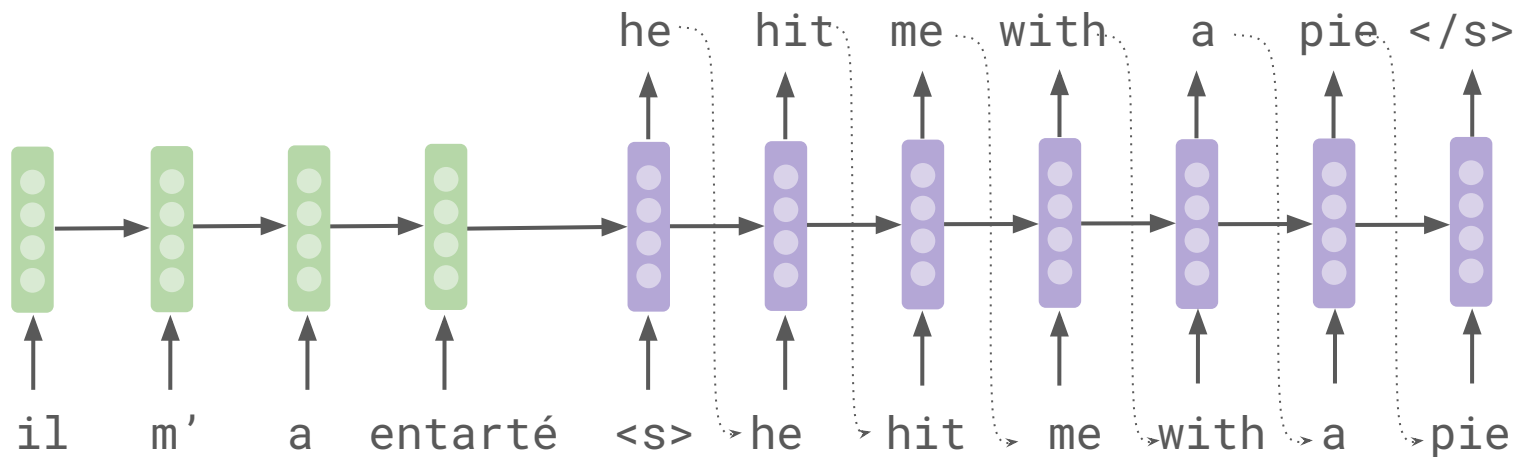
Neural machine translation



Encoder RNN inputs **word embeddings** for the source language

Decoder RNN inputs **word embeddings** for the target language

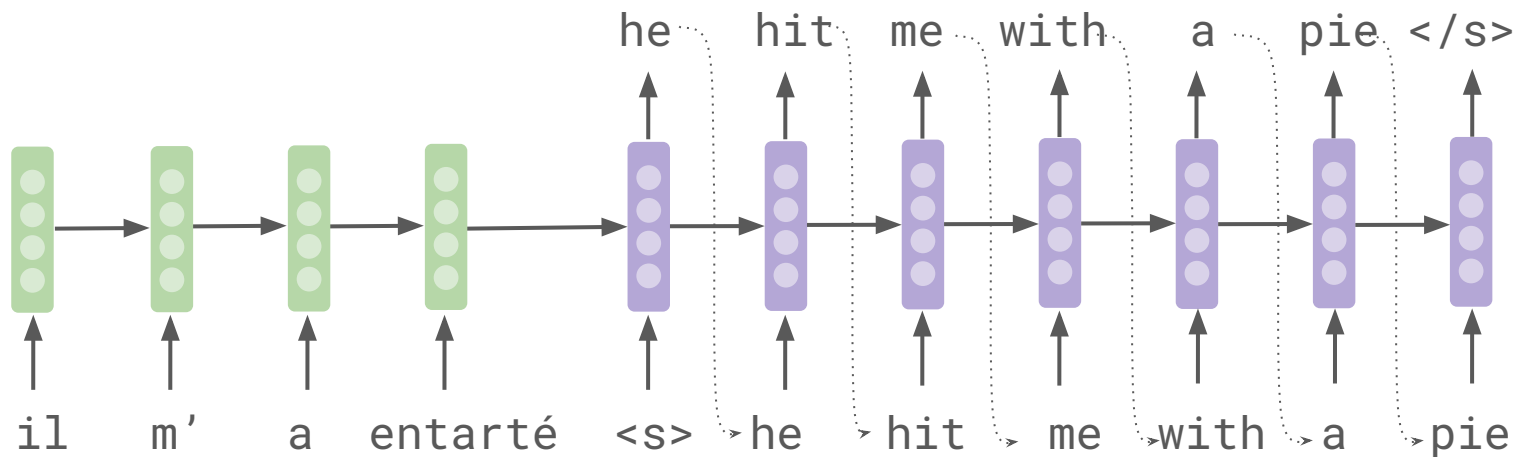
Neural machine translation



Encoder RNN has a set of parameters e.g. $\mathbf{W}$, $\mathbf{U}$, $\mathbf{b}$.

Decoder RNN has a separate set of parameters e.g. $\mathbf{W}'$, $\mathbf{U}'$, $\mathbf{b}'$.

Neural machine translation



Encoder RNN can be
bidirectional

Decoder RNN has to be unidirectional

Neural machine translation

The seq2seq model is an example of a **conditional language model**

- Language model because the decoder is predicting the next word of the target sentence y
- Conditional because its predictions are also conditioned on the source sentence x

$$P(y|x) = P(y_1|x)P(y_2|y_1x) \dots P(y_T|y_1, \dots, y_{T-1}, x)$$

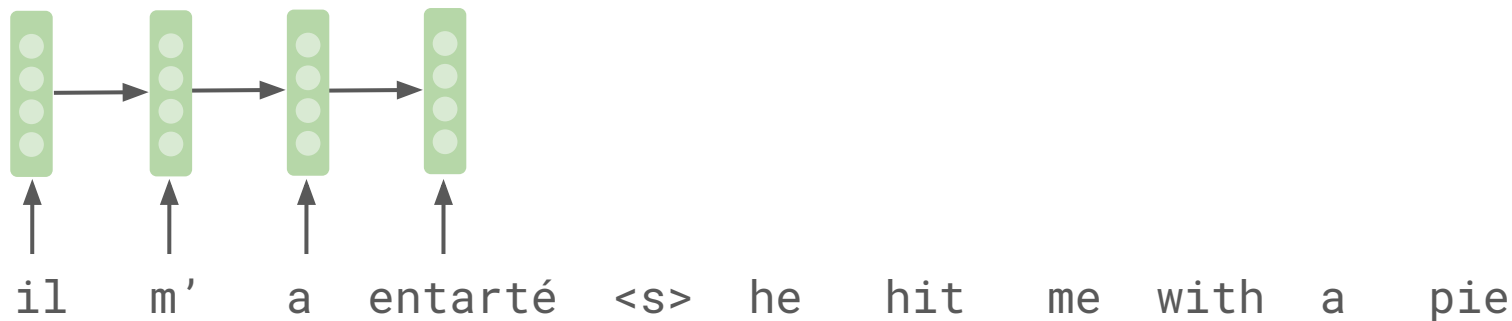
Probability of the next target word, given target words so far and source sentence x

Training NMT

il m' a entarté <s> he hit me with a pie

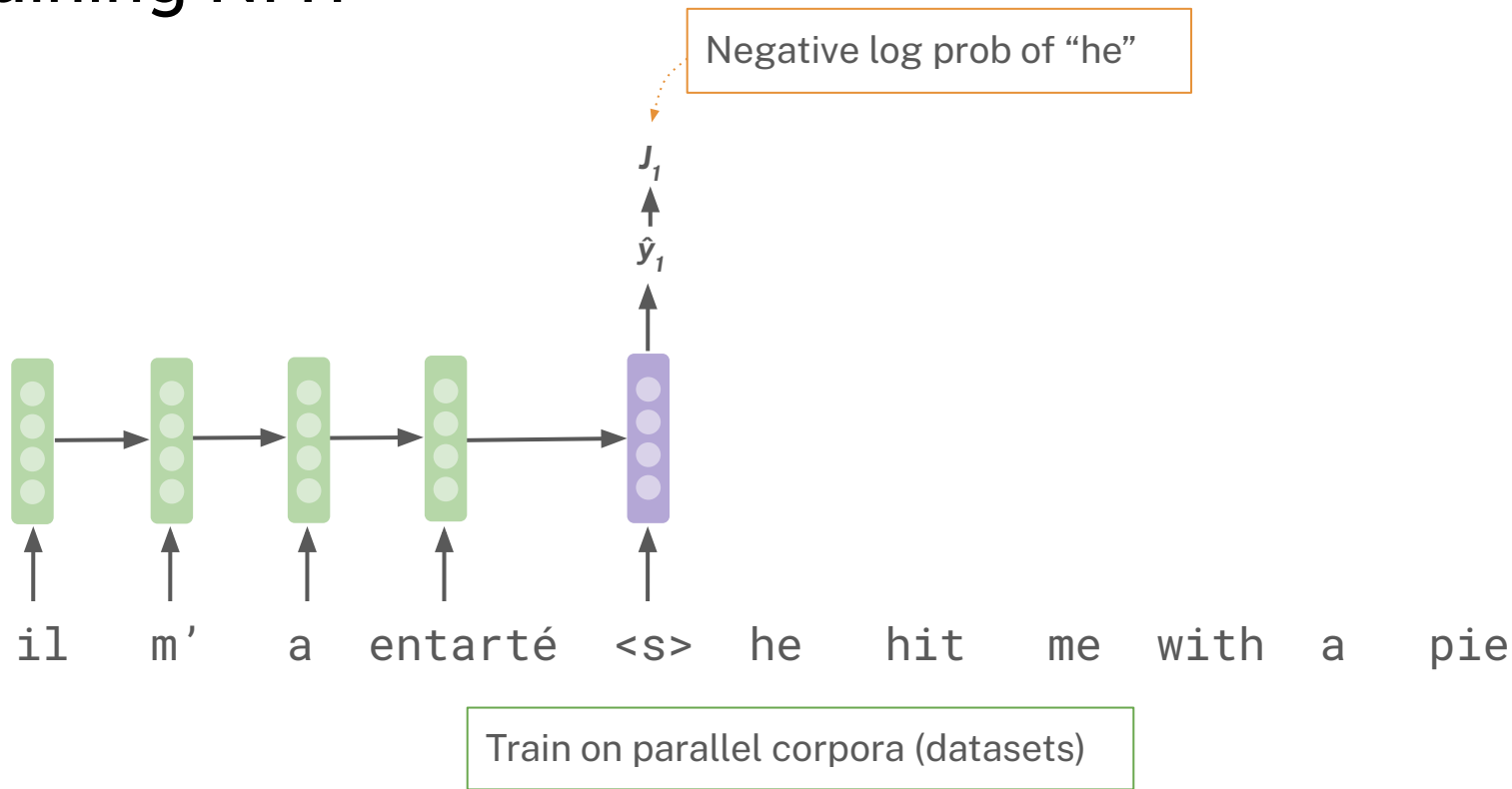
Train on parallel corpora (datasets)

Training NMT

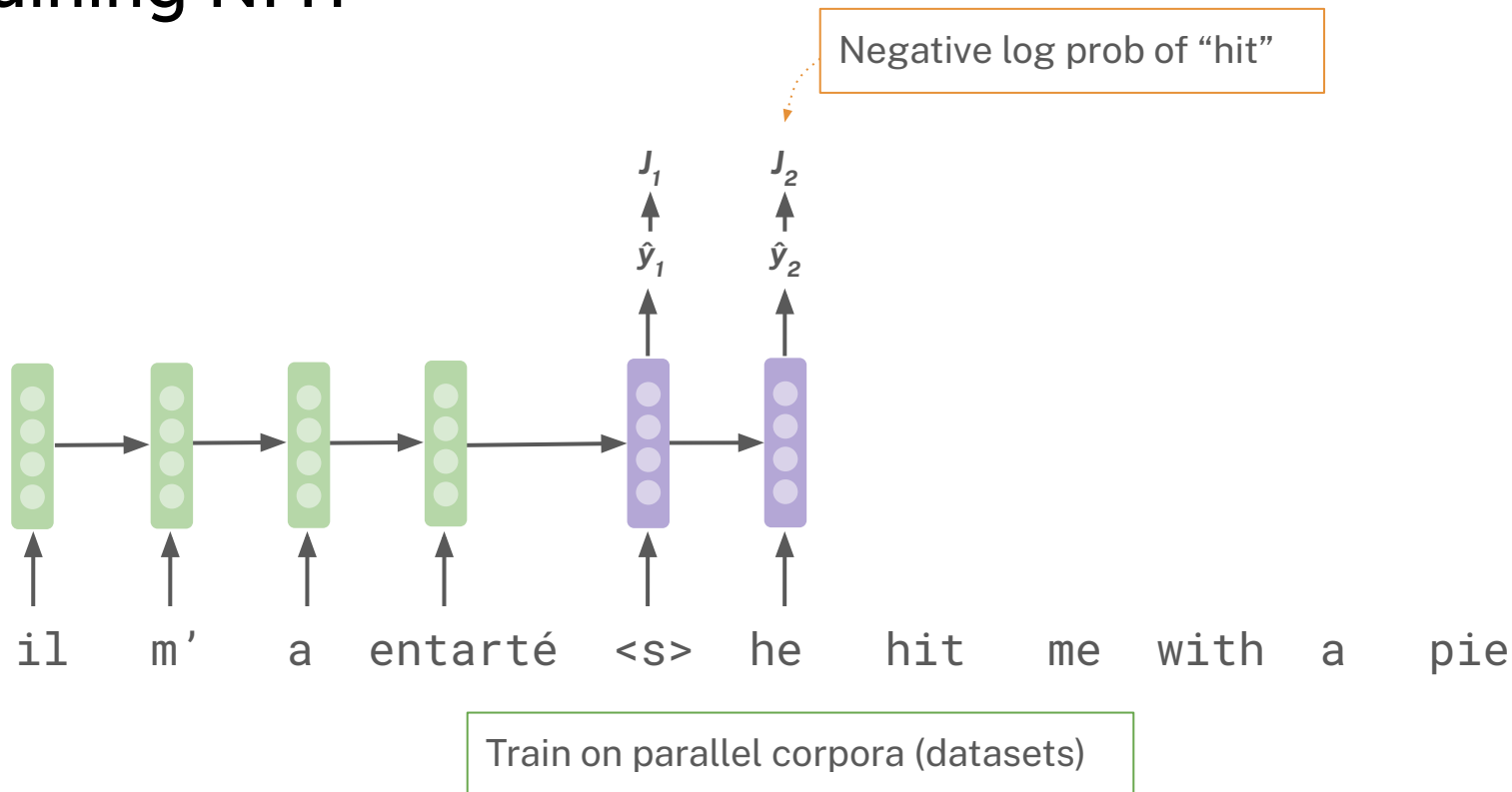


Train on parallel corpora (datasets)

Training NMT

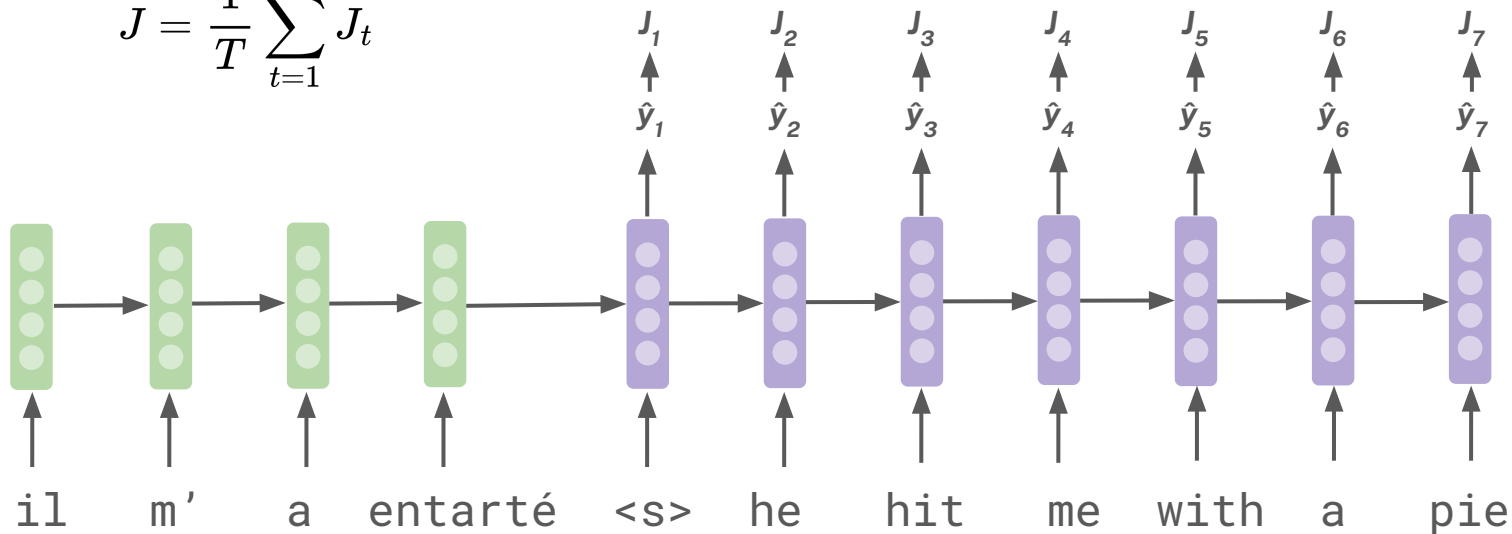


Training NMT



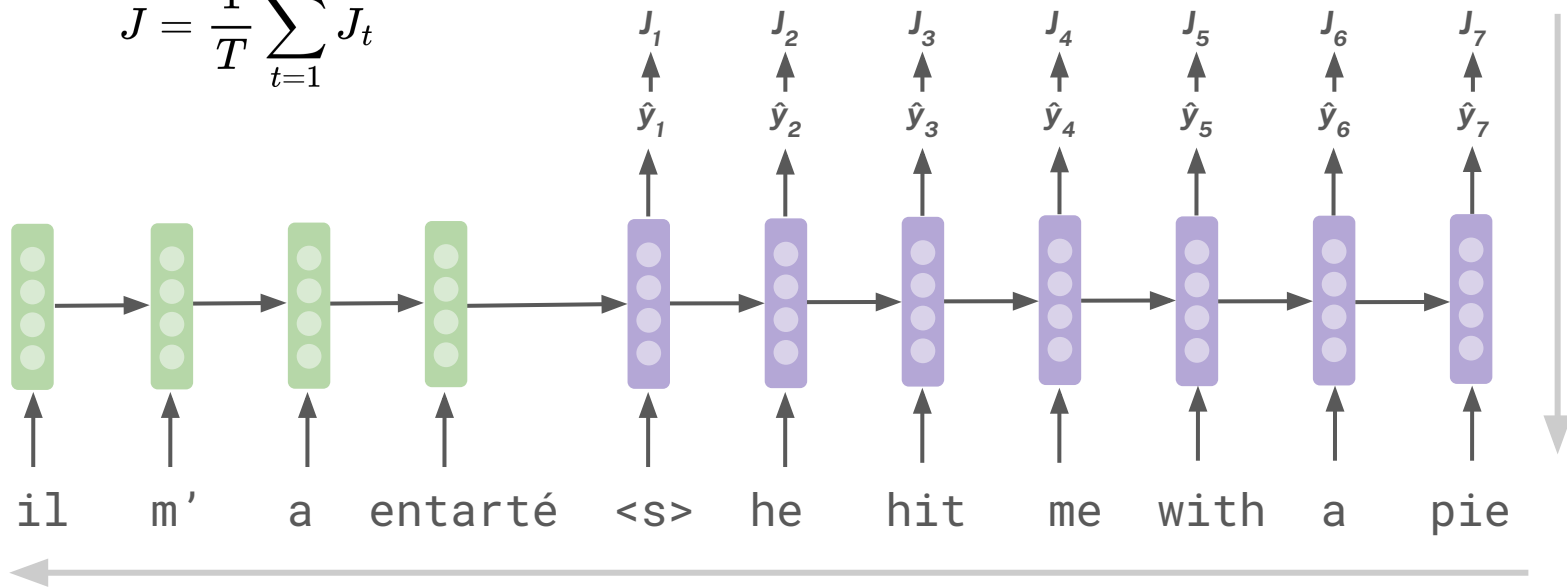
Training NMT

$$J = \frac{1}{T} \sum_{t=1}^T J_t$$



Training NMT

$$J = \frac{1}{T} \sum_{t=1}^T J_t$$



Seq2seq is optimized as **a single system**. Backprop operates end-to-end.

What if we don't have enough data?

Data augmentation using monolingual LMs and bilingual lexicons



A boy is holding a bat.

computer vision
→
augmentation



A boy is holding a bat.

Plausible substitutes are suggested by a monolingual LM:

My sister drives a [car / motorbike] ✓

My uncle sold his [house / motorbike] ✓

Alice waters the [plant / motorbike] ✗

John bought two [shirts / motorbike] ✗

Prioritize including examples involving **rarer words**.

A boy is holding a bat. translation
Ein Junge hält einen Schläger. → A boy is holding a **backpack**.
 augmentation Ein Junge hält einen *Rucksack*.

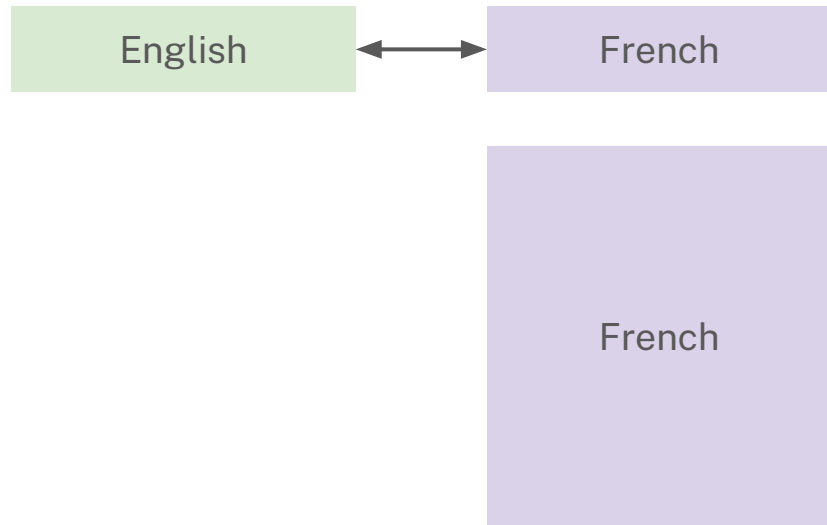
[Fadaee et al. 2017](#)

Provided by a bilingual lexicon

What if we don't have enough data?

Back translation

Given a small parallel dataset b/t source (e.g. English) and target (e.g. French) + a huge monolingual target dataset



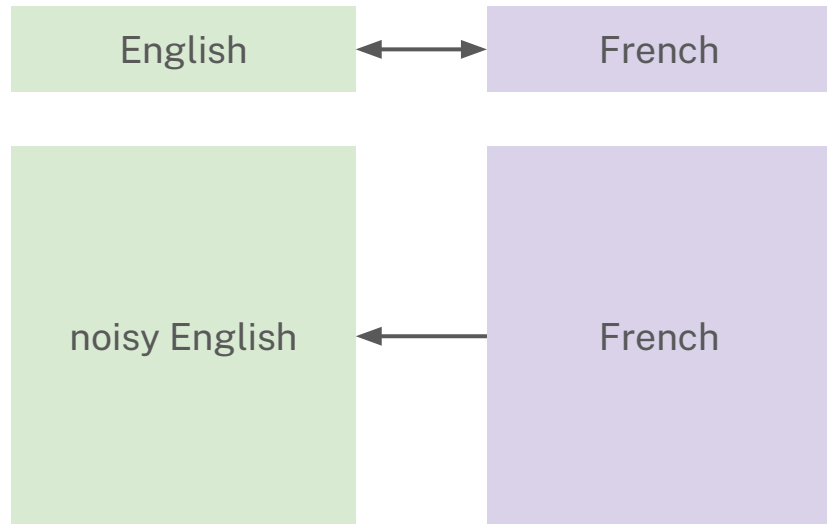
What if we don't have enough data?

Back translation

Given a small parallel dataset b/t source (e.g. English) and target (e.g. French) + a huge monolingual target dataset

Train a target $\rightarrow$ source model

Use this model to translate target monolingual corpus $\rightarrow$ source lang



What if we don't have enough data?

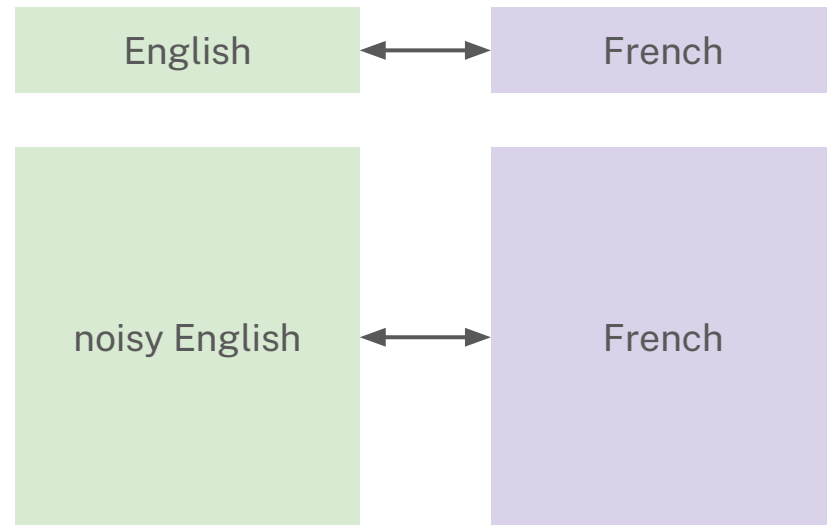
Back translation

Given a small parallel dataset b/t source (e.g. English) and target (e.g. French) + a huge monolingual target dataset

Train a target $\rightarrow$ source model

Use this model to translate target monolingual corpus $\rightarrow$ source lang

Use all of this data to train a **source** $\rightarrow$ **target** model!



Seq2seq is versatile

Sequence-to-sequence is useful for more than just MT

Many NLP tasks can be phrased as sequence-to-sequence:

- Summarization (long text → short text)
- Dialogue (previous utterances → next utterance)
- Parsing (input text → output parse as sequence)
- Code generation (natural language → Python code)

MT in the LLM era

Team	Approach
Algharb	LLM
AMI	LLM
COILD-BHO	LLM
CommandA-WMT	LLM
CUNI-Doc Transformer	enc-dec
CUNI-EdUKate-v1	LLM
CUNI-SFT	LLM
CUNI-Transformer	enc-dec
DLUT_GTCOM	LLM, enc-dec
HYT	LLM
GemTrans	LLM
In2x	LLM
NTTSU	LLM
SalamandraTA	LLM
SH	LLM
Shy-hunyuan-MT	LLM
Systran	LLM
TranssionMT	enc-dec, LLM
UvA-MT	LLM
Wenyiil	LLM
Yandex	LLM
Yolu	LLM

[Kocmi et al. 2025](#)

Though LLMs dominate the MT landscape now, challenges like **data quality and size** still remain.

Another key challenge has also stuck around...

Metrics for **evaluating** MT, e.g. BLEU, COMET, BLEURT:

- may penalize valid lexical choices and phrase reorderings that don't match the ground-truth reference
- may delude us into thinking a translation is “high quality” but still miss significant grammatical errors

Much discussion around how to *best* evaluate translations!

We'll talk about stuff like this more in our evaluation & benchmarking lecture later.

Resources and competitions for MT

 Machine Translate

[More](#) / [Associations](#) / [WMT](#)

WMT

The Conference on Machine Translation

WMT is the main event for machine translation and machine translation research. The conference is held annually in connection with larger conferences on natural language processing.

The conference aims to bring together academic scientists, researchers and industry representatives to exchange and share their experiences and research results. WMT plays a key role for the entire industry of computational linguistics and machine translation.

Events

Calls for papers

Translation APIs

Models

Features

Input modes

Integrations

Routers

Quality estimation

Automatic post-editing

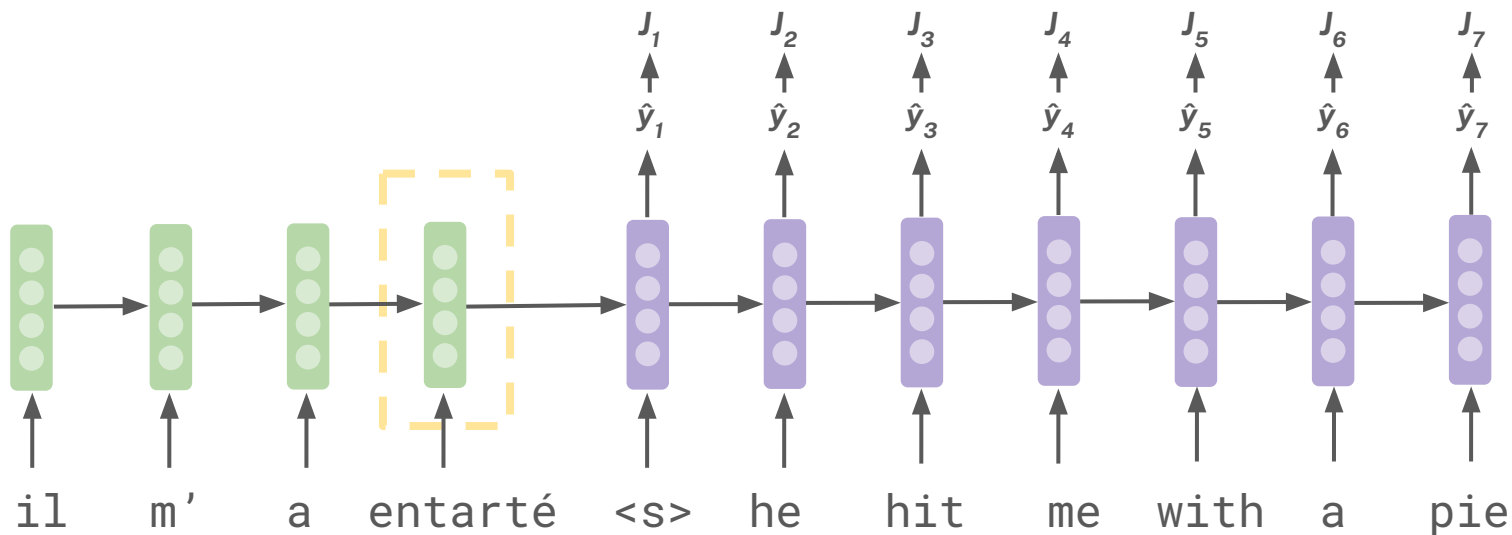
Languages

Foreshadowing

The bottleneck

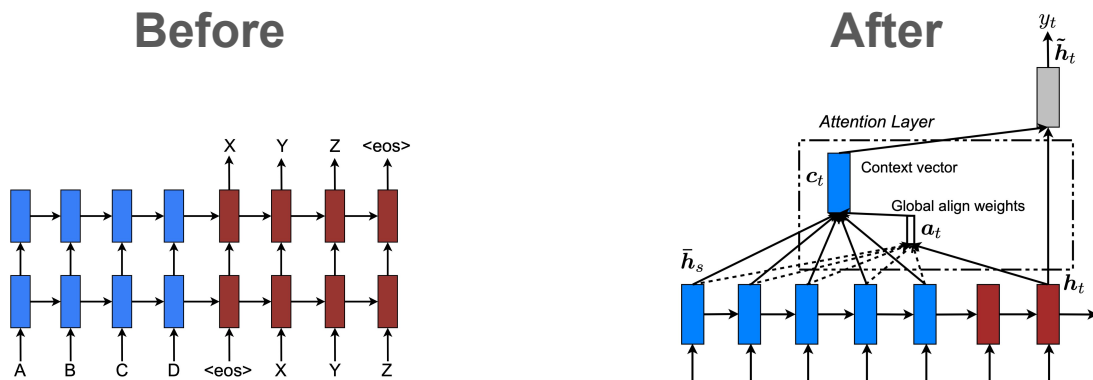
One encoding vector needs to capture all of the info about the source sentence.

- + Longer sentences can lead to vanishing gradients!



Attention

Attention provides a solution to the bottleneck problem



Next class!!

What we covered today

1. Language modeling (e.g. n -gram LMs, window-based neural LMs)
2. Recurrent neural networks (RNNs)
3. Machine translation (a very language-y task involving RNNs)

Please form your project groups by EOD!

Last time I checked there were ~6 groups

If you don't yet have a group, come up to me during the remaining class time and we can figure out group match-ups right now, right here.

Otherwise I will assign you groups.

Upcoming milestone

Project proposal presentation (due Sept 22)

- Time per presentation TBD; will announce in class and on course website “Assignments” page after all groups are formed.
- This **live pitch** should include why the question/s being asked are important and why your project can be completed feasibly within this semester. Grading rubric will be finalized by next class.
- Each project team will be assigned to me or Sonia as the “main” mentor based on the presentation pitch.
- It’s okay to modify your topic between your brainstorming pitch and your actual written proposal, but if you do this, explicitly say you did